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Protecting PII Data in LLM's by using Encoding/ Introducing noise at client end

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Final-Thesis Report

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I would like to thank all the people that helped me in my thesis, ¹UpGrad & LJMU mentors, online platform and discussion forums for the information and support in perusing my thesis. Next, I want to thank my family for extending their support throughout the time, and I want to thank my Mentor Sushrut and Student Mentor Rubina for their quick responses and helping in balancing my work and study. Also special thanks to my friends Varunjai, Manoj, Prateek and colleagues in office and Upgrad.

Abstract

There has been ever growing concerns about the privacy in this world of Large language models(LLM) where anyone chatting with AI chatbots is skeptical about how his/her information will be leaked out or can be mis-used. This paper investigates various privacy-preserving methods for client data protection in Large Language Model inference. There are already many privacies preserving chatbots in market but they either lack appropriate privacy preservation or are very computational heavy. This paper examined existing approaches including the Split-N-Denoise framework and cryptographic techniques such as homomorphic encryption and various perturbation methods to analyze how they balance utility and privacy, with particular focus on local differential privacy applications for text data.

This paper proposes the CleanSplit Model (CSM) a novel privacy-preserving inference framework that deploys only a lightweight token representation layer on the client side. The primary objective of CSM is to provide a framework for picking right method to protect user privacy in client side chatbots that uses LLM inference, thereby optimizing the critical trade-off between privacy preservation and computational utility.

This paper evaluated various techniques to protect private data particularly PII (Personally Identifiable Information) using techniques like Faking PII data, Masking PII data, replacing PII data using client side LLM. Evaluation used various Metrics likes PII Leakage Rate, BLEU Score, ROUGE Scores (ROUGE-1, ROUGE-2, ROUGE-L) etc to check cohesivity of the outputs from LLM.

As an outcome our approach demonstrated that depending upon the context of problem someone has to pick right privacy protection solutions as an example, if user is using chatbot where an immediate answer is required Faking is most appropriate technique with modest PII leakage rate and low latency whereas if use case is Batch based or used in ultra sensitive domains, client side LLM are more powerful in protecting PII data with very less PII leakage rate and little high latency.

Finally, CSM proposed a framework that helps to pick right solution to protect PII data to balance efficiency and result quality.

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LIST OF ABBREVIATIONS

LLM	Large Language Models
GDPR.....	General Data Protection Regulation
HIPAA.....	Health Insurance Portability and Accountability Act
PII	Personal identifiable information
NIST	National Institute of Standards and Technology
SMPC	Secure Multi-Party Computation
HE	Homomorphic Encryption
SnD	Split-N-Denoise
CSM.....	CleanSplit Model
NER	Named Entity Recognition
ML.....	Machine Learning
HaS.....	Hide and Seek
DP.....	Differential Privacy
DUR.....	Data Utility Rate
PHI.....	Protected Health Information
PIB.....	Personally Identifiable Behaviour

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CHAPTER 1

INTRODUCTION

The growing concern over data privacy in Large Language Models (LLMs) is reflected in recent governmental and regulatory interventions, including the EU's AI Act, the US Executive Order on Safe, Secure, and Trustworthy AI, and established regulations such as **GDPR and HIPAA**. These frameworks emphasize the necessity for AI systems to incorporate robust privacy-preserving mechanisms, making the development of practical and efficient solutions for local privacy protection in LLM inference paramount.

1.1 Background of the Study

The escalating threat of data breaches is evident across multiple sectors, especially in healthcare and financial services, where sensitive personal identifiable information (PII) and monetary assets are at stake.

PII (Personally Identifiable Information) refers to any data that can directly or indirectly identify a person or individual. It is a foundational concept in data privacy, cybersecurity, and regulatory compliance — especially relevant when studying data breaches in financial institutions. According to the U.S. National Institute of Standards and Technology (NIST SP800-122): PII is “any information about an individual maintained by an agency, including any information that can be used to distinguish or trace an individual’s identity.”

As per latest HIPPA report (Steve Alder, 2025) in June 2025, the healthcare sector witnessed a 16.67% monthly increase in security incidents, and a 302.71% rise in individuals affected by PHI (Protected Health Information) breaches compared to the previous month(May). Tens of millions of individuals’ PII are compromised monthly, with over 7.6 million affected in just June 2025. Globally, cybercrime's economic impact is projected to reach US\$10.5 trillion by 2025, growing about 15% annually.

Financial institutions are similarly under intense pressure and exposure. As per Journal article (Olumayowa Adefowope Adekoya, 2025) nearly 46% of financial institutions worldwide reported experiencing a data breach in the last 24 months. In 2024, about

65% of financial organizations reported ransomware attacks—up from 64% in 2023 and dramatically higher than 34% in 2021. The finance sector is currently the most affected industry under GDPR in the UK: from 2023 through Q1 2025, finance businesses self-reported 2,175 breaches.

The increased reliance on digital services comes with an expanded cyber security attack surface for malicious actors. This expanded attack surface has led to a rise in cyber-attacks. Consequently, the global average cost of a data breach increased by 10 % to 4.88 million dollars in 2024, according to IBM's Data Breach Cost Report (IBM Corporation, 2025). As per report, the cost per data breach for financial firms is among the highest of all industries. For example, in 2024, for large-scale breaches (50 million records or more compromised), financial firms faced average costs of US\$375 million. Detection and containment times for financial firms are also significant, it took about 168 days on average to detect a breach, and 51 days to contain it. The volume of records compromised in the financial sector has soared. Between 2022 and 2023, records affected increased from ~29.3 million to over 43.6 million breached records across financial companies.

In terms of incident frequency, in the first half of 2025, financial services reported 387 million data breaches in the United States, making it one of the most breached industries. These figures underscore that financial organizations do not merely face more attacks; the stakes are higher in terms of cost, scale, duration of exposure, and potential downstream harm (fraud, reputational damage, regulatory penalties). Together with the healthcare data, they paint a picture of sectors under acute threat — reinforcing the necessity of advanced defence mechanisms, robust detection/prevention, and rapid response strategies. Thus, there is a need to develop a framework to protect PII data and comply with regulations.

1.2 Problem Statement

The rapid adoption of large language models (LLMs) across sensitive domains such as healthcare, finance, and education has heightened concerns around data privacy, particularly regarding the protection of personally identifiable information (PII). While several privacy-preserving techniques have been proposed, existing methods continue to face critical limitations that hinder their effectiveness in real-world deployments.

Figure 1.1 explains that when using client side chatbots how PII data in customer prompts might be leaked to server side LLM and can be memorized by LLM's for subsequent replies.

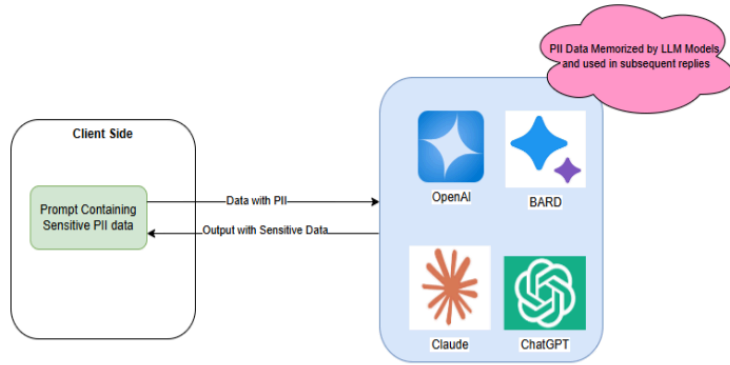


Figure 1.1 No PII Protection

Cryptographic approaches—such as Secure Multi-Party Computation (SMPC) and Homomorphic Encryption (HE)—offer strong theoretical guarantees for data privacy. However, they impose significant computational overhead, resulting in increased latency and resource consumption. This makes them unsuitable for time-sensitive applications that demand real-time or near-real-time inference, such as clinical decision support systems or financial transaction monitoring.

Alternatively, perturbation-based methods, including those relying on differential privacy, attempt to obfuscate sensitive information by adding statistical noise. While these methods can improve scalability, they often struggle to maintain an optimal balance between data utility and privacy. The trade-off frequently leads to diminished model performance, reduced accuracy, and unreliable outputs—especially detrimental in high-stakes environments where precision is critical.

Figure 1.2 explains when using privacy layer like SMPC, HE, DP in client side chatbots how still some PII data in customer prompts might be leaked to server side LLM (Due to inefficiencies with those techniques discussed later in the paper in Chapter 2) and can be memorized by LLM's for subsequent replies.

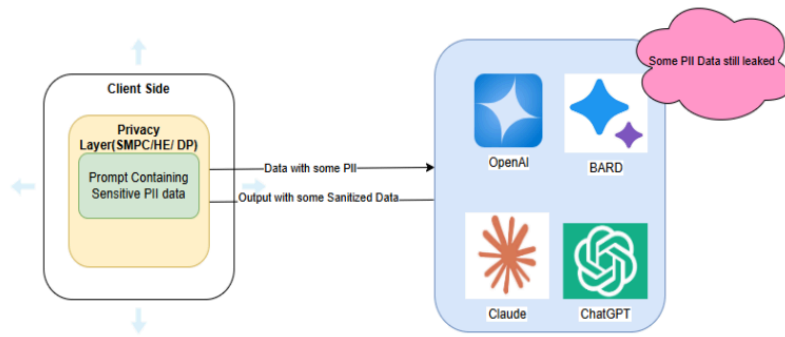


Figure 1.2 PII Data Protection using SMPC/HE/DP techniques

Recently, the Split-N-Denoise (SnD) framework has emerged as a hybrid solution that introduces differential privacy at the architectural level. While it offers improvements in utility preservation compared to pure perturbation methods, SnD is not without its flaws. Notably, the injection of noise before transmission to LLM servers can still result in indirect PII leakage, especially if the noise does not fully obfuscate identifiable tokens. Moreover, the absence of integrated Named Entity Recognition (NER) mechanisms means that many identifiable entities—such as names, locations, and financial identifiers—can pass through unrecognized and unprotected. This poses a significant privacy risk, particularly in sectors governed by stringent regulations such as HIPAA, GDPR, and financial compliance frameworks.

Given these limitations, there is a pressing need for a more robust and practical solution that ensures strong privacy guarantees without sacrificing model performance or scalability. This research aims to address this gap by proposing the CleanSplit Model (CSM)—a novel client-side privacy-preserving inference framework. CSM incorporates a NER-driven approach to detect and replace sensitive entities prior to model interaction, ensuring that no raw PII is ever transmitted to LLM servers. Unlike cryptographic or noise-based techniques, CSM seeks to maintain computational efficiency while achieving a higher level of privacy assurance.

By focusing on real-world applications in healthcare, finance, and education—where the protection of PII is not only a legal requirement but a foundational element of user trust, this study contributes a practical, efficient, and secure framework for private LLM inference. The proposed solution is intended to bridge the gap between privacy, utility,

and usability, ultimately enabling safer and more responsible use of AI in sensitive domains.

Figure 1.3 explains when using privacy layer of CSM Framework in client side chatbots how PII data in customer prompts is protected and not leaked to server side LLM.

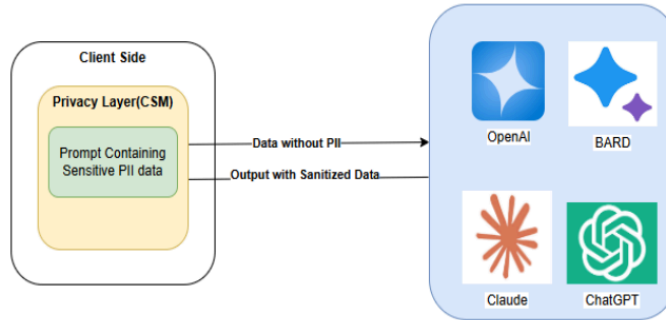


Figure 1.3 PII Data with CSM Privacy Layer

1.3 Aim and Objectives

Aim:

The intent of this paper is to develop a novel architectural framework that helps to pick right methodology for efficiently protecting personally identifiable information (PII) in Large Language Model inference while maintaining computational efficiency, semantic context, and model performance in privacy-sensitive applications.

Objectives:

Objective 1: Comprehensive Analysis of Existing Privacy-Preserving Methods.

- a) **To systematically** review existing privacy-preserving techniques for LLM inference, including cryptographic protocols (e.g., homomorphic encryption, secure multiparty computation) and differential privacy frameworks.
- b) **To identify** specific gaps and vulnerabilities in existing privacy-preserving techniques that hinder real-time deployment in sensitive domains such as healthcare and finance.

Objective 2: Development of Client-Side PII Detection and Replacement Framework.

- a) **To design** a client-side Named Entity Recognition (NER)-based model for identifying Personally Identifiable Information (PII) in natural language inputs.
- b) **To develop** an algorithmic strategy for generating contextually relevant synthetic replacements that preserve semantic and conversational coherence.
- c) **To assess** the accuracy, latency, and robustness of the PII detection and replacement framework under varying data conditions and model architectures.

Objective 3: Provide Framework to help in picking right methodology for balanced Utility and Performance Optimization.

- a) **To propose** a modular architectural framework integrating the PII protection layer into the LLM inference pipeline.
- b) **To implement** optimization techniques (e.g., model quantization, lightweight NER architectures, and efficient token handling) to enhance client-side performance.
- c) **To benchmark** the proposed architecture various techniques used (NER replacement, Masking, local LLM replacement for PII) in terms of response latency, detection accuracy, and resource utilization.

1.4 Research Questions (IF ANY)

This study addresses three fundamental research questions that are critical to developing an effective privacy-preserving LLM inference framework:

This study seeks to address the following research questions, which are central to the development of a technically robust and privacy-preserving LLM inference framework.

RQ1: What are the problems in existing PII protection mechanisms and how efficiently can NER models be deployed client-side while minimizing overhead/latency?

To what extent can Named Entity Recognition (NER) models be efficiently deployed on the client side to extract PII entities while minimizing computational overhead and latency in real-time chatbot applications? This question investigates the trade-offs between lightweight, rule-based, and transformer-based NER approaches in terms of detection accuracy, processing speed, and device resource utilization.

RQ2: How does PII substitution influence the accuracy, coherence, and semantic integrity of responses?

How does the substitution of detected PII entities with synthetic or contextually equivalent placeholders influence the accuracy, coherence, and semantic integrity of LLM-generated responses? This research question focuses on quantifying performance variations arising from data anonymization and identifying optimal replacement strategies that preserve conversational naturalness.

RQ3: How can privacy be safeguarded while retaining contextual continuity in multi-turn conversations?

How can user privacy be fully safeguarded while retaining contextual continuity and coherence across multi-turn conversational sessions? This question explores techniques for context reconstruction, anonymization consistency, and memory handling to achieve compliance with domain-specific privacy requirements (e.g., in healthcare and financial services) without degrading model usability.

Table 1.1 provides how research questions are lined to objectives and gives justifications for the alignments as well.

Table 1.1 Mapping of Research Questions to Objectives

Research Question (RQ)	Linked Objectives	Description of Alignment
RQ1: What are the problems in existing PII protection mechanisms and how efficiently can NER models be deployed client-side while minimizing overhead/latency?	1.a, 1.b, 2.a, 2.c, 3.b	These objectives collectively explore and evaluate existing privacy-preserving methods, design a lightweight client-side NER mechanism, and optimize computational performance to minimize latency and resource usage in real-time applications.
RQ2: How does PII substitution influence the accuracy, coherence, and semantic integrity of responses?	2.b, 2.c, 3.c	These objectives focus on designing and testing PII replacement algorithms, quantifying the effect on LLM inference accuracy and semantic coherence, and analyzing trade-offs between anonymization strength and conversational quality.
RQ3: How can privacy be safeguarded while retaining contextual	1.3, 3.1, 3.3	These objectives address methods for preserving conversation flow while ensuring complete privacy protection, through architecture design, experimental validation, and

continuity in multi-turn conversations?		formulation of deployment guidelines for privacy-compliant conversational systems.
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1.5 ¹ Scope of the Study

In this case study, the following points cover the scope of the study:

- Study existing Models
- Explain the Dataset used
- Explain Pre-processing done
- Finding out solution that doesn't compromise on Privacy and Speed
- Propose Architecture only covers PII data (like Contact details, Demographic information, Personal identifiers etc) protection and efficiency
- Study doesn't cover other type of secret data like:
 - PHI (Protected Health Information) - Medical conditions, prescriptions, diagnoses.
 - PIB (Personally Identifiable Behaviour)- Browsing history, app usage, search queries.
 - Secrets or Credentials- API keys, passwords, access tokens, SSH keys.
 - Financial Information - Bank balances, investment data, transaction details.
 - Intellectual Property (IP) - Source code, internal documentation, trade secrets.
 - National Security or Classified Data - Defence-related terms, classified project names.

1.6 Significance of the Study

Data breach costs hit a record peak of \$4.88 million in 2024, representing a 10% rise from the previous year. Approximately 46% of security incidents compromised customer personally identifiable information, encompassing tax identification numbers, email addresses, telephone numbers, and residential addresses. In 2024, ~275 million healthcare records were breached in the U.S (2024-healthcare-data-breach-report).

Financial institutions face an average data breach cost of roughly \$6.08 million per incident when personally identifiable information is compromised in 2025. Regulatory penalties for non-compliance may total up to 4% of worldwide annual revenue under GDPR, with other frameworks imposing thousands of dollars per individual violation. Approximately 50% of U.S. citizens have suffered personally identifiable information breaches within the past five years, as billions of data records face annual exposure. These financial calculations exclude difficult-to-measure consequences such as psychological trauma, credit rating deterioration, and missed opportunities stemming from PII misuse.

Proposed Architecture that could enhance the protection of PII data from client side chatbots whereby providing a more reliable and efficient way of preserving PII data. This would enhance the security and trust of LLM chatbots particularly amongst financial/ healthcare and education industry where protecting your PII data is utmost important and is mandated by regulators. This contributes in potentially reducing frauds by not leaking PII data over internet.

1.7 Structure of the Study

This research is organized into six comprehensive chapters, each addressing a key aspect of the investigation into privacy-preserving mechanisms for Large Language Models (LLMs) using PII-aware processing frameworks.

Chapter 1 – Introduction

This chapter introduces the research background, motivation, and problem statement. It highlights the growing importance of privacy protection in AI-driven text generation, emphasizing risks associated with Personally Identifiable Information (PII) leakage. The chapter defines research objectives, hypotheses, and key contributions, and outlines the scope and limitations of the study.

Chapter 2 – Literature Review

The literature review surveys existing research on PII detection, anonymization, and data protection frameworks. It analyzes current trends in data privacy laws (e.g., GDPR, HIPAA), traditional masking techniques, and recent advancements in LLM-based PII

handling. Comparative studies of existing anonymization and obfuscation methods are examined to identify knowledge gaps and motivate the proposed approach.

Chapter 3 – Research Methodology

This chapter details the research design, methodology, and experimental framework adopted for the study. It defines the overall approach, including dataset selection, preprocessing steps, evaluation metrics, and implementation tools. The conceptual design of the CleanSplit Model or equivalent privacy-preserving framework is elaborated, supported by system architecture diagrams and processing pipelines.

Chapter 4 –Implementation

This chapter provides a comprehensive overview of the dataset used (e.g., Kaggle PII dataset), including data characteristics, structure, and annotation procedures. It explains data preprocessing, labeling of PII entities, and integration of the dataset into the experimental pipeline. Implementation details of each anonymization strategy—Real Names (Baseline), Fake Names Replacement, XXXX Masking, and LLM-based PII Removal—are discussed with examples and configuration parameters.

Chapter 5 – Results and Evaluation

This chapter presents the empirical analysis and quantitative results. It covers:

Evaluation Metrics: Privacy, quality, and efficiency indicators such as PII Leakage Rate, Re-identification Risk, BLEU, ROUGE, Perplexity, and Coherence.

Experimental Procedure: Data preparation, execution pipeline, and statistical testing.

Results and Discussion: Comparative analysis of each method’s privacy protection effectiveness, linguistic quality, and efficiency trade-offs.

Key findings are interpreted in the context of privacy–utility balance and their implications for secure LLM deployment.

Chapter 6 – Conclusion and Future Work

The final chapter summarizes major findings, theoretical and practical contributions, and limitations of the study. It concludes with recommendations for applying privacy-preserving frameworks in real-world AI systems and proposes future directions, such as

adaptive anonymization, federated PII learning, and integration of regulatory compliance mechanisms.

CHAPTER 2

LITERATURE REVIEW

Large Language Models (LLMs) and other advanced machine learning (ML) systems have become integral to countless applications, from personal assistants to healthcare diagnostics (Yu et al., 2022) (Sun et al., 2025) (Asthana et al., 2025). Their power is derived from training on vast datasets, which frequently contain Personally Identifiable Information (PII) and other sensitive data. This dependency creates a significant privacy risk, as ML models have been shown to memorize and leak sensitive information from their training data, both during the training process and at inference time when interacting with users. These leakages can be exploited through various attacks, leading to breaches of individual privacy and exposing organizations to legal and reputational damage under regulations like GDPR and HIPAA (Lukas et al., 2023) (Johnson et al., 2020) (Asthana et al., 2025).

This review surveys the current landscape of PII data protection in machine learning. It begins by exploring the fundamental nature of privacy in the context of natural language and outlines the primary threats and attack vectors that lead to PII leakage. It then provides a comprehensive overview of the major categories of privacy-preserving techniques, including data-level sanitization, model-level and algorithmic defences like Differential Privacy and distributed learning, and cryptographic methods such as Homomorphic Encryption. Finally, it discusses the methodologies for evaluating these techniques and concludes by summarizing the key challenges and future research directions in this critical field.

2.1 Introduction

The proliferation of Large Language Models (LLMs) has led to their widespread adoption in conversational agents and chatbots across sensitive domains such as healthcare and customer service. While LLMs offer powerful capabilities, they inherently introduce critical privacy and security concerns. These concerns stem primarily from the models' tendency to memorize and regurgitate specific phrases, passages, or entire sensitive records from the data they process.

Privacy risks in chatbots extend beyond training data leakage to the inference stage, where user prompts themselves may contain Personally Identifiable Information (PII) or Protected Health Information (PHI). This sensitive input can be involuntarily retained by the model or actively exploited through various attack vectors, including training data extraction, PII extraction, reconstruction, and inference attacks. This body of work seeks to evaluate and compare strategies for mitigating these risks by transforming user input before it reaches the conversational AI, balancing the competing goals of *privacy protection* and *utility preservation*.

Research has demonstrated that LLM-powered chatbots, such as ChatGPT, are "highly prone to retaining and copying PII from the input when prompted". In one study, when prompted to summarize cover letters containing PII, the model retained PII verbatim in over half of the cases (57.4%). This involuntary leakage can occur even when the model is not malicious, simply due to its nature of memorizing data (Priyanshu et al., 2023)

Many studies have been conducted for coming up with a solution to anatomize the user inputs that contain PII sensitive data.

Differential Privacy

According to (Yang et al., 2022) (Wei et al., 2020) (Yu et al., 2022) (Kerrigan et al., 2020), the implementation of a *generalized gradient perturbation split learning framework* introduces a robust mechanism for ensuring differential privacy within distributed and federated learning environments. Their studies collectively emphasize the necessity of preserving data confidentiality when multiple clients collaboratively train a shared model without exposing their local data. In this framework, gradient updates transmitted between local clients and the central server are perturbed with statistically calibrated noise. This technique provides formal (ϵ, δ) -differential privacy guarantees, ensuring that the presence or absence of any single user's data has a negligible influence on the overall model parameters or communication exchanges.

While this approach significantly enhances privacy protection, it introduces a well-documented privacy-utility trade-off. As observed by (Yang et al., 2022) and (Kerrigan et al., 2020), achieving stronger privacy (represented by lower epsilon values) necessitates injecting higher magnitudes of noise into the gradient signals. This process inevitably degrades model accuracy, slows convergence, and reduces the overall

predictive reliability of the trained model. The impact of this degradation becomes even more pronounced when differential privacy is applied to small-scale or sparse datasets, where noise can obscure meaningful patterns to such an extent that the model's analytical capability is severely compromised (Wei et al., 2020).

In the context of chatbot and conversational AI systems, these limitations are particularly restrictive. Chatbots often rely on relatively small, domain-specific, or personalized dialogue datasets, where linguistic nuances and contextual continuity are critical to system performance. The excessive noise introduced by differential privacy mechanisms can distort semantic representations, leading to incoherent responses, loss of contextual relevance, and reduced conversational fluency. Consequently, although differential privacy-based gradient perturbation frameworks are theoretically sound and highly suitable for large-scale systems such as financial or healthcare networks, their practical applicability to dialogue-driven AI systems remains limited.

To overcome these challenges, recent research has begun exploring alternative privacy-preserving strategies that maintain a better balance between confidentiality and model performance. These include federated learning with selective gradient sharing, knowledge distillation-based privacy, and secure multi-party computation (SMPC).

Homomorphic Encryption Protection Analysis

Research by (Gilbert & Gilbert, 2024),(Bae et al., 2025), (Liu & Liu, 2023) has highlighted Homomorphic Encryption (HE) as a promising cryptographic approach for ensuring robust data privacy in distributed and cloud-based computing environments. Unlike conventional encryption schemes that require data to be decrypted before computation, HE enables direct mathematical operations on encrypted data while preserving the confidentiality of the underlying information. This means that intermediate systems—such as cloud servers or third-party processors—can perform complex computations (e.g., addition, multiplication, or model inference) on ciphertexts without ever accessing the plaintext. As a result, sensitive datasets such as personally identifiable information (PII) or financial records remain fully encrypted throughout the entire data processing lifecycle, significantly reducing exposure risk and mitigating potential attack vectors.

Despite these strong theoretical advantages, homomorphic encryption faces substantial practical limitations that hinder its widespread adoption in real-world applications. The most significant challenge is its computational inefficiency. As noted by(Bae et al.,

2025), operations performed on encrypted data can be up to 1,000 times slower than equivalent operations on plaintext. This high computational overhead stems from the mathematical complexity of maintaining algebraic consistency across encrypted domains, often requiring intensive modular arithmetic and large ciphertext sizes. Consequently, HE-based systems demand specialized hardware accelerators, high memory capacity, and substantial energy consumption to achieve acceptable performance levels.

These constraints make real-time processing of sensitive data—such as continuous financial transactions, credit scoring, or medical monitoring—largely impractical without significant architectural optimization. Furthermore, implementing HE at scale introduces additional design complexities related to key management, ciphertext expansion, and error accumulation, which can further degrade system performance. Therefore, while homomorphic encryption provides strong theoretical privacy guarantees, its current computational burden limits its practicality for latency-sensitive applications like banking analytics, fraud detection, or conversational AI systems handling PII. Ongoing research aims to mitigate these issues through partially homomorphic schemes, hardware-assisted acceleration (e.g., GPUs, FPGAs), and hybrid models that combine HE with other privacy-preserving techniques such as secure multi-party computation (SMPC) or differential privacy to achieve a more balanced trade-off between security and efficiency.

Hide and Seek (HaS)

(Y. Chen et al., 2023) introduced an innovative framework aimed at safeguarding prompt privacy within large language models (LLMs), addressing one of the most pressing concerns in the deployment of conversational AI systems—unintended exposure of sensitive or personally identifiable information (PII) embedded within user prompts. The proposed HaS (Hide and Seek) system integrates two complementary methodologies: concealment of private entities through anonymization and retrieval of those entities via controlled de-anonymization. In the anonymization phase, the framework employs specialized components to identify and mask sensitive entities (such as names, account numbers, or identifiers) before the prompt is processed by the LLM. Subsequently, in the de-anonymization phase, the concealed entities can be selectively restored once the model's output has been verified as privacy-safe, ensuring that private information remains protected throughout the inference process.

The HaS architecture employs a generative Bloomz model—a multilingual open-source large language model—trained on data annotated by GPT-4, which provides high contextual sensitivity and entity-recognition accuracy. This dual-model setup enables dynamic anonymization and de-anonymization during real-time interaction, effectively reducing the risk of PII leakage through prompts or model responses. However, despite its conceptual robustness, the framework faces significant practical limitations. The incorporation of dual processing components and the reliance on large generative models substantially increase computational overhead on the client side, making chatbot deployments resource-intensive. Moreover, the latency introduced by the two-stage anonymization–de-anonymization pipeline hampers real-time conversational performance, particularly in low-power or latency-sensitive environments such as mobile and enterprise customer service applications. Consequently, while the HaS framework demonstrates strong potential for privacy preservation in LLM-driven systems, it currently struggles to achieve the necessary balance between privacy assurance, computational efficiency, and user experience, highlighting the need for lightweight yet effective prompt privacy mechanisms in production-grade chatbot architectures.

Split-N-Denoise Framework Analysis

(Mai et al., 2024) (Vepakomma et al., 2018) proposed a privacy-preserving inference framework that introduces a client-side denoising model to improve data utility following noise-based perturbation. The central premise of this approach lies in leveraging prior knowledge of raw input characteristics and the corresponding noise distribution to reconstruct more informative embeddings after the application of privacy-preserving noise. By performing denoising at the client level, the framework seeks to retain the functional accuracy of downstream tasks such as classification or prediction, even when strong Differential Privacy (DP) constraints are imposed during the data transmission or aggregation process. This hybrid strategy attempts to balance the competing objectives of maintaining privacy protection and ensuring model usability across distributed machine learning environments.

However, despite its innovative architecture, this framework exhibits critical limitations in safeguarding Personally Identifiable Information (PII). The core of the approach still depends heavily on Differential Privacy mechanisms, which are fundamentally designed to protect *statistical aggregates or summary information* rather than *individual*

data instances. As highlighted by (Vepakomma et al., 2018) and later reaffirmed by (Mai et al., 2024), the injected noise in DP-based systems can reduce the probability of identifying individuals but does not entirely eliminate the risk of re-identification or membership inference attacks, especially when adversaries possess auxiliary information.

This vulnerability becomes particularly concerning in high-stakes domains such as healthcare and finance, where regulations—such as HIPAA, GDPR, or the Digital Personal Data Protection (DPDP) Act—require complete protection or anonymization of PII. In such environments, even partial exposure of sensitive attributes (e.g., transaction IDs, patient identifiers, or demographic information) can have severe legal and ethical implications. Moreover, when applied to sensitive textual data, the stochastic noise introduced through DP can distort semantic content without fully obfuscating identifiable linguistic patterns, making de-anonymization theoretically possible. Consequently, while client-side denoising enhances the utility of differentially private embeddings, it fails to achieve comprehensive PII protection, underscoring the need for more robust hybrid privacy frameworks that combine DP with encryption-based or secure multi-party computation (SMPC) techniques to ensure both privacy and functional fidelity.

The Nature of Privacy and PII in Language Models

Protecting PII in language models is more complex than simply redacting direct identifiers like names or social security numbers as described in paper by (Lukas et al., 2023). The sources highlight a fundamental tension between technical data protection methods and the inherently social and contextual nature of privacy.

A core challenge is that "data protection is not equivalent to privacy protection". True privacy preservation requires respecting contextual integrity—understanding the norms and context in which information is shared, an ability current LLMs lack. Humans respect privacy not by failing to memorize information, but by judging the appropriateness of sharing it, a nuance that technical systems cannot yet replicate (Brown et al., 2022). Many privacy-preserving techniques are based on assumptions that do not hold for natural language, where private information can be amorphous, embedded in prose, and shared among groups of individuals, blurring the boundaries of data ownership and sensitivity.

Furthermore, the very definition of a "leak" is complex. While many studies focus on verbatim memorization example by (Ippolito, 2023), where a model exactly reproduces a string from its training set, this is an insufficient standard. Research shows that models can easily be prompted to produce approximate memorization—paraphrased or stylistically altered versions of sensitive text, which bypasses simple verbatim filters, creating a false sense of privacy. This suggests that a significant portion of privacy risks may be underestimated by current evaluation methods.

2.2 Privacy Threats and Attack Vectors

The leakage of PII can occur unintentionally through model behaviour or be actively exploited by malicious adversaries. The literature categorizes these threats into several key types of privacy attacks, some of them are discussed in detail in paper by (Jegorova et al., 2022).

Involuntary Leakage: Involuntary leakage refers to the unintentional exposure of Personally Identifiable Information (PII) that occurs as a natural consequence of how machine-learning models learn from data. This form of leakage does not rely on malicious intent; instead, it emerges because large neural networks tend to memorise rare or unique patterns in the training dataset. Several studies have shown that large language models can unintentionally reproduce sensitive training data segments. For instance, (Carlini et al., 2021) demonstrated that GPT-based models can regurgitate memorised sequences such as names, phone numbers, and email addresses under specific prompting conditions.

Leakage may also occur indirectly via internal representations. (de Amorim et al., 2020) showed that deep models can embed identifiable patterns that allow sensitive attributes to be inferred through representation analysis. Similarly, (Salem et al., 2018) demonstrated that even when models are accessible only through APIs, carefully crafted queries can extract sensitive information or reconstruct parts of the training data. These findings show that leakage is not limited to explicit memorization but may also arise from representational learning.

Training choices also influence the extent of involuntary leakage. According to ar (Feldman, 2021), overparameterized models tend to memorize outliers more frequently, making leakage an inherent characteristic of modern deep networks.

Overall, the literature recognizes involuntary leakage as a **structural privacy limitation** of current machine-learning paradigms. Even when data is not explicitly stored, the learning process may encode enough information for reconstruction or inference. These challenges motivate the need for robust mitigation strategies—such as differential privacy, regularization, and strong data-governance practices—discussed later in this thesis as part of the proposed CSM framework.

Memorization and Regurgitation: LMs are prone to memorizing and reproducing sequences from their training data, particularly atypical or frequently repeated data points, which are more likely to be sensitive as pointed in papers by (Brown et al., 2022)(Jegorova et al., 2022). This risk extends to the inference stage, where models may copy, or regurgitate, PII from a user's input prompt directly into the output. Studies show that this regurgitation can be frequent and may even exhibit biases across different demographic subgroups (Priyanshu et al., 2023).

Malicious Attacks: Malicious attacks refer to deliberate attempts by adversaries to extract sensitive information from trained machine-learning models. Unlike involuntary leakage, which arises passively from the learning process, malicious attacks exploit model vulnerabilities through carefully constructed queries, reverse-engineering techniques, or gradient-based manipulation. The literature identifies several attack categories, each demonstrating that privacy risks persist even when direct training data access is restricted.

One of the most widely studied malicious attack vectors is the membership inference attack, where an adversary attempts to determine whether a particular data point was part of the model's training set.

Another major class involves model inversion attacks, where the adversary reconstructs sensitive attributes or input samples by exploiting the model's output space. Later studies, such as (Carlini et al., 2021), improved the reconstruction quality and showed that inversion is possible even in black-box conditions.

A related line of research explores training data extraction attacks, in which adversaries reconstruct verbatim sequences or structured personal data through crafted prompts or perturbations. Large language models have shown notable vulnerability in this respect. (Carlini et al., 2021) demonstrated that GPT-like systems can reveal memorized phone numbers, names, addresses, and other sensitive text fragments when prompted

adversarial. These results highlight that malicious probing can trigger behavior not ordinarily observable during normal operation.

Additionally, property inference attacks, as studied by (Melis et al., 2018), allow an adversary to infer global properties of the training data—such as demographics or distribution characteristics—even without reconstructing individual records. Such attacks present risks in contexts involving protected or regulated attributes.

Collectively, these findings underscore that malicious adversaries can exploit even well-trained and seemingly secure models to extract confidential information. This evidence motivates the need for privacy-preserving mechanisms—such as differential privacy, regularization-based defenses, and controlled access policies—which are discussed in later chapters under the proposed CSM framework.

Data Extraction Attacks: An attacker with API access can query a model to reconstruct verbatim or approximate training examples. These can be untargeted, aiming to recover any training data, or targeted, focusing on specific types of PII like email addresses or phone numbers as described in papers by (Brown et al., 2022)(Huang et al., 2023)° (Lukas et al., 2023)

Reconstruction and Inference Attacks: A more informed attacker can use a sentence with masked PII (e.g., "John Doe lives in [MASK]") to have the model reconstruct the missing information. This is even more effective if the attacker has a list of potential candidates (PII inference). Advanced attacks can leverage both the prefix and suffix of the masked context to improve success rates by over 10x compared to simpler methods, same technique has been discussed in paper (Lukas et al., 2023)

Indirect Prompt Injection: This attack exploits a fundamental flaw in LLMs, their inability to reliably separate instructions from data. An attacker can embed malicious commands within a piece of text meant for processing (e.g., an email to be summarized). The LLM may then execute this hidden command, potentially leading it to leak PII or perform other unauthorized actions as described in paper (Zverev et al., 2025)

This section outlines how privacy risks in Large Language Models arise from both unintentional model behavior and deliberate adversarial attacks. It highlights key leakage pathways including memorization, regurgitation, and targeted extraction; as well as sophisticated inference and prompt-injection techniques that exploit model

vulnerabilities. Together, these threats underscore the need for robust safeguards to prevent exposure of sensitive PII.

2.3 Privacy-Preserving Techniques for PII Protection

A diverse array of techniques has been developed to mitigate PII leakage. These can be broadly categorized into data-level, model-level, and cryptographic approaches. Some of these are discussed in paper (Jegorova et al., 2022)

Some of these techniques are as discussed.

Data-Level Techniques: Sanitization and Anonymization: Data-level privacy techniques focus on modifying the raw input data before it is processed by machine-learning models, ensuring that sensitive information is removed, obfuscated, or transformed in a manner that prevents re-identification. These approaches operate as a pre-processing layer, offering protection independent of the downstream model’s architecture or behavior. Sanitization methods typically involve detecting and deleting explicit identifiers such as names, email addresses, and phone numbers, while anonymization extends further by replacing sensitive elements with structurally consistent but non-identifying substitutes.

One of the most common strategies in this category is Named Entity Recognition (NER)-based de-identification, where automated models detect PII entities and mask or replace them before further processing. Tools based on BERT, spaCy, or domain-specific NER systems are frequently used for this purpose; however, existing literature points out that these models often suffer from limited recall, domain dependence, and degraded performance on datasets with unseen entity types or inconsistent annotation schemes (Johnson et al., 2020). Consequently, some sensitive items may remain unmasked even after automated sanitization.

Another widely studied technique is token-level masking, where sensitive terms are substituted with generic placeholders (e.g., *[NAME]*, *[ADDRESS]*). While effective in preventing leakage, such masking can distort sentence structure and diminish the utility of downstream natural-language tasks, especially those requiring semantic continuity or contextual inference.

More advanced anonymization methods employ synthetic replacement, such as fake-name generation or surrogate keys, which preserve linguistic fluency and coherence. Studies such as (Ji Young Lee et al., 2017) demonstrate that synthetic substitution strikes

a more favorable balance between privacy and utility, particularly in conversational datasets.

Recent work explores LLM-based redaction, where large language models themselves are used to rewrite or sanitize text while retaining the semantic intent. Although these approaches can deliver high linguistic quality, they introduce new risks related to hallucination, inconsistency, and the possibility of the model reintroducing sensitive information if not rigorously constrained (Priyanshu et al., 2023).

Overall, data-level sanitization and anonymization serve as foundational mechanisms for privacy preservation because they decouple risk mitigation from model inference. Nonetheless, their effectiveness relies heavily on the accuracy of PII detection, the consistency of anonymization rules, and the trade-off between privacy strength and linguistic utility.

NER-Based De-identification: A common approach is to use Named Entity Recognition (NER) models, such as fine-tuned BERT, to identify and scrub PII from text. However, these models are also imperfect; Existing NER models have limited recall, meaning some PII is inevitably missed, and performance degrades significantly when applied to new data domains due to differing annotation standards and data distributions. These limitations are discussed in (Lukas et al., 2023)(Johnson et al., 2020) .

Furthermore, recent works emphasize the need for **context-aware** de-identification, arguing that NER alone is insufficient for complex privacy decisions that require semantic understanding—such as distinguishing between personal identifiers and generic references with similar structure (Parker et al., 2022). As a result, emerging research proposes hybrid pipelines that combine NER with rule-based post-processing, LLM-driven semantic filtering, or probabilistic privacy scoring.

These persistent limitations underscore the difficulty of achieving fully reliable PII removal using NER alone. In Chapter 3, this study discusses these challenges in detail and introduces enhancements aimed at improving recall, domain robustness, and contextual accuracy within NER-driven de-identification pipelines.

Client-Side Prompt Sanitization: For API-based LLMs where the model itself cannot be modified, sanitization can be performed on the client's device before the prompt is sent to the server. Frameworks like PP-TS (Kan et al., 2023), Casper (Chong et al.,

2024) , and Rescriber (Zhou et al., 2025) implement this. They range from fully automated multi-layered systems combining rules, NER, and local LLMs for topic detection (Casper), to user-in-the-loop tools that empower users to make final decisions on redaction or abstraction (Rescriber). These tools enhance user control and can have an educational effect by raising privacy awareness. A key challenge is ensuring sanitization does not compromise the utility of the LLM's response, which some frameworks address with "reasonability checks" (Kan et al., 2023).

This section reviews major privacy-preserving strategies designed to prevent PII leakage in LLM systems, spanning data preprocessing, client-side sanitization, and broader anonymization techniques. It highlights the strengths and limitations of NER-based de-identification and discusses emerging client-side frameworks that balance privacy protection with task utility. Collectively, these approaches demonstrate both progress and persistent challenges in safeguarding sensitive information during model interaction.

2.3.1 Algorithmic Techniques: Distributed Learning and Differential Privacy

These techniques modify the model training or inference process to protect privacy.

Distributed Learning: The core idea is to keep raw data decentralized on user devices. Federated Learning (FL) enables multiple clients to collaboratively train a model without sharing their local data; only model updates (e.g., gradients or weights) are sent to a central server for aggregation as discussed in (Wei et al., 2020). Combining FL with Parameter-Efficient Fine-Tuning (PEFT), such as LoRA, makes it feasible to tune large models on resource-constrained devices, a paradigm known as Federated Instruction Tuning (FedIT) (Zhang et al., 2024). In this setting, the heterogeneity of data across clients becomes an advantage, improving the robustness and generalizability of the global model.

Split Learning (SplitNN) splits the neural network architecture between the client and the server. The client processes data through the initial layers and sends only the intermediate activations to the server, which completes the computation (Vepakomma et al., 2018). This approach can be more computationally and communication-efficient for the client than FL in certain scenarios, such as when model sizes or the number of clients is large (Singh et al., 2019). Privacy can be further enhanced by adding noise to

the intermediate activations, with frameworks like Split-N-Denoise (SnD) using a client-side denoising model to recover utility from the server's noisy output.

Differential Privacy (DP): DP is the "gold standard" for providing mathematically rigorous privacy guarantees by adding calibrated noise to data or computations (Li et al., 2022)(Kumar et al., 2024). DP-SGD is the primary algorithm for training neural networks with DP, which involves clipping the norm of per-example gradients and adding noise before aggregation (Yu et al., 2022) (Jegorova et al., 2022).

Applying DP to large models is challenging due to high computational costs and utility degradation. However, combining DP-SGD with PEFT methods like LoRA has proven highly effective for privately fine-tuning LLMs. This approach significantly reduces overhead and can lead to private models whose performance approaches that of their non-private counterparts. Surprisingly, studies show that larger models can be more robust to DP noise and experience a smaller relative drop in accuracy when fine-tuned this way.

Despite its strong guarantees, DP is not a panacea. Its record-level privacy definition is a poor fit for interconnected textual data, and it has been shown to still permit a non-trivial amount of PII leakage, especially for information that appears frequently across the dataset.

This subsection surveys algorithmic privacy-preserving methods—primarily federated and split learning, along with differential privacy—that modify training workflows to protect user data. It highlights how decentralized learning reduces raw data exposure and how DP techniques provide formal privacy guarantees despite practical challenges in scalability and model utility. Together, these approaches offer promising yet imperfect solutions for securing sensitive information during model training.

2.3.2 Cryptographic Techniques

Homomorphic Encryption (HE) HE enables computation directly on encrypted data, offering a powerful way to outsource processing to untrusted servers without revealing plaintext information (Gilbert & Gilbert, 2024a) .

HE for Private Inference: A major challenge is that HE schemes typically only support polynomial operations (addition and multiplication), while transformers contain complex non-polynomial functions like GELU, softmax, and Layer-Norm(T. Chen et

al., 2022). The **THE-X** framework addresses this by approximating these functions with HE-friendly alternatives (e.g., replacing GELU with ReLU) and using a hybrid client-server protocol where the client performs the non-polynomial steps on temporarily decrypted data.

HE for Private Retrieval: For retrieval-augmented LMs, HE can be used to build an encrypted vector database. This allows a client to perform secure semantic search over a private, remote database, retrieving relevant documents without the server learning anything about the query or the data content (Bae et al., 2025b).

The primary limitation of HE is its significant performance overhead in computation and communication, which currently restricts its use in many real-time applications (Gilbert & Gilbert, 2024a).

This subsection examines cryptographic approaches—particularly Homomorphic Encryption—that enable computation on encrypted data to protect sensitive information during inference or retrieval. While HE offers strong confidentiality guarantees and supports private model execution and secure semantic search, its practical adoption is limited by substantial computational and communication overhead. These methods demonstrate high theoretical security but remain constrained for real-time or large-scale LLM applications.

2.4 Evaluation Methodologies and the Privacy-Utility Trade-off

Evaluating the effectiveness of PII protection techniques requires balancing two competing goals: privacy and utility. A central finding across the literature is the existence of a privacy-utility trade-off: stronger privacy protections, such as adding more noise in DP, generally lead to a degradation in model performance (Yu et al., 2022).

To systematically measure this trade-off, dedicated evaluation frameworks have been proposed. The "Privacy-preserving4LLM Benchmarking" framework, for instance, provides a methodology for assessing algorithms across different real-world scenarios (e.g., protecting training data vs. user queries) and uses a Parameter Optimizer to ensure fair comparisons (Sun et al., 2025).

Privacy Metrics: Privacy is quantified using several methods. The exposure metric involves inserting "canary" data—unique, sensitive sequences—into a dataset and measuring how easily they can be extracted from the trained model (Sun et al., 2025). Other methods rely on the success rate of the attacks described in Section 3, such as the number of PII successfully extracted or the accuracy of reconstruction attacks (Huang et al., 2023).

Utility Metrics: Utility is typically measured using standard NLP evaluation metrics on downstream tasks, such as perplexity for language modelling, accuracy and F1-score for classification, and BLEU or ROUGE scores for text generation (Kerrigan et al., 2020).

Intuitive approaches, such as anonymizing sensitive terms before LLM input, prove insufficient for comprehensive privacy protection. These methods fail to conceal other linguistic elements vital for semantic interpretation, thereby compromising both privacy and the effectiveness of tasks requiring precise semantic understanding. Additionally, existing denoising techniques face fundamental limitations when deployed server-side, as the server lacks knowledge of injected noise levels, creating an inherent conflict with privacy protection objectives.

This section discusses how evaluating PII protection techniques requires navigating the inherent privacy–utility trade-off, where stronger safeguards often reduce model performance. It reviews emerging benchmarking frameworks, key privacy and utility metrics, and highlights why seemingly simple approaches like anonymization or server-side denoising fail to provide comprehensive protection. Overall, the evaluation landscape underscores the difficulty of achieving both strong privacy and high-quality LLM utility.

2.5 Research Gap and Contribution

The literature reveals a persistent and critical gap in the development of privacy-preserving methodologies capable of simultaneously ensuring robust protection of individual PII and maintaining the semantic integrity and performance of large language models (LLMs). Existing approaches—such as those based on Differential Privacy, Homomorphic Encryption, or client-side denoising frameworks—tend to exhibit a trade-off between privacy strength and data utility. Methods offering stronger privacy guarantees often degrade model accuracy and fluency, while utility-preserving

techniques fail to provide sufficient protection for fine-grained, individual-level information. Consequently, current privacy-preserving solutions remain inadequate for high-sensitivity applications, particularly in sectors like healthcare and financial services, where conversational AI systems process context-rich, personally identifiable data under stringent regulatory constraints (e.g., GDPR, HIPAA).

Another problem that exists is how to choose between different types of methodologies to protect PII data as some of them doesn't preserve utility and some of the methods doesn't provide performance, leading to the dilemma to pick and choose right method for right use case.

This research advances the understanding of privacy-preserving language model operations through several significant contributions. It introduces a structured, modular evaluation framework for assessing the impact of privacy-preserving transformations on LLM outputs, enabling reproducibility and comparative analysis. A multi-dimensional metric design—integrating BLEU, ROUGE, and Coherence—captures lexical, semantic, and contextual quality more comprehensively than single-metric approaches. The study further provides empirical insights into the performance of masking, fake-name substitution, and LLM-based redaction, quantifying their influence on text quality and coherence.

To address these limitations, this study proposes an enhanced architecture termed the CleanSplit Model (CSM). The CSM framework introduces a hybrid privacy-preserving mechanism that integrates Named Entity Recognition (NER) with context-aware data synthesis. In this architecture, NER algorithms are employed to detect and classify PII entities—such as names, contact details, or financial identifiers—within user inputs. Identified entities are then replaced with synthetically generated, semantically aligned placeholders, ensuring that the contextual flow and intent of the conversation remain intact. Unlike traditional anonymization or noise-based approaches, CSM preserves both linguistic coherence and domain relevance, thereby minimizing the trade-off between privacy protection and model performance.

Also, CSM provides a framework that helps to pick the right methodology for right use case, some of the domains need high privacy over performance and some of the domains might need more performance over privacy or some need a right balance between the two.

By maintaining contextual fidelity while guaranteeing complete PII masking, the CleanSplit Model represents a promising direction for developing trustworthy, privacy-

compliant chatbot systems. Its design directly supports deployment in regulatory-intensive domains, such as healthcare diagnostics, financial advisory services, and digital customer support—where safeguarding user trust and adhering to compliance standards are as crucial as model accuracy and responsiveness.

Finally, the study contributes a reusable benchmark dataset and methodology, and demonstrates the applicability of client-side NER and CSM in real-time chatbot environments, supporting privacy-compliant deployment in regulated sectors such as healthcare, finance, and customer support.

2.6 Conclusion and Future Directions

The protection of PII in machine learning is a complex and rapidly evolving field. This review has highlighted the multifaceted nature of privacy threats, from unintentional model memorization to sophisticated adversarial attacks, and surveyed the three main pillars of defence: data-level sanitization, algorithmic approaches like DP and distributed learning, and cryptographic methods like HE.

While significant progress has been made, no single technique offers a complete solution. Data sanitization is practical but often superficial and brittle. Distributed learning architectures like FL and SplitNN are powerful for keeping data local but still expose model updates or activations that can leak information. Differential Privacy offers provable guarantees but comes at a cost to utility and is based on assumptions that are a poor fit for natural language. Homomorphic Encryption provides strong security but remains computationally expensive for widespread use.

The most promising solutions appear to be hybrid approaches. Parameter-efficient fine-tuning has made DP practical for large models. Client-side tools that combine multiple sanitization techniques and empower users with control are gaining traction. Cryptography is finding a role in securing specific components of larger systems, like retrieval databases.

Key challenges remain. The community must move beyond simplistic, verbatim definitions of memorization to address approximate or semantic leakage (Ippolito, 2023). The fundamental lack of instruction-data separation in LLMs poses a systemic risk that may require architectural innovations to solve (Zverev et al., 2025). Finally, bridging the gap between technical data protection and the rich, contextual nature of

human privacy, performance remains the ultimate goal and a central open question for future research (Brown et al., 2022).

Another challenge that exists is how to pick the right methodology for right use case to protect PII data and balance utility and performance.

Table 2.1 provides a comprehensive list of papers used in literature review along with the methodology used and outcome that paper has achieved. Table also specifies the shortcomings and future research that can be done.

Table 2.1 Various papers used in Literature review starting from the latest first

Paper	Methodology Used	Outcome	Shortcomings / Future Research
(Bae et al., 2025).	The methodology employs a hybrid framework combining Socratic Chain-of-Thought Reasoning with a Homomorphically Encrypted Vector Database	The hybrid setup of remote and local LLMs enhances privacy and boosts performance	Future work should target tasks resistant to clean decomposition,
(Sun et al., 2025)	The framework includes a Protection Scenario Manager, a Parameter Optimizer for utility-privacy balance, and a Privacy-Level Evaluator using canary exposure, analyzing three DP algorithms across multiple LLMs and datasets.	CusText+ offers optimal utility-privacy balance, Mahalanobis ensures strongest privacy, Llama2-7b is compatible, Mistral-7B resilient, and Falcon-7b vulnerable.	Current privacy-preserving algorithms trade utility for protection, prompting future work on tailored attacks, algorithms, and latency effects
(Zhou et al., 2025)	Rescriber is a browser extension using on-device LLMs for user-led data minimization via detection, sanitization, and privacy recovery, evaluated with 12 users comparing Llama3-8B and	Rescriber empowered user-driven data minimization and showed Llama3-8B performs comparably to GPT-4o in usability tests.	The study highlights privacy misperceptions, abstraction gaps, and utility loss, urging better small LLMs and placeholder strategies.

	GPT-4o against Presidio.		
(Zverev et al., 2025)	The study defines and measures instruction-data separation in LLMs using an empirical separation score with "surprise witnesses," evaluated on the SEP dataset to assess mitigation techniques like prompt engineering and fine-tuning.	Experiments show LLMs lack clear instruction-data separation; while prompt engineering and fine-tuning improve separation, they reduce utility, suggesting the need for architectural changes for robust mitigation.	The paper notes limitations in using proxy prompts and highlights persistent failures even in advanced LLMs; future work calls for new architectures and training methods to achieve true instruction-data separation and improved explainability.
(Mai et al., 2024)	The Split-N-Denoise (SnD) method safeguards privacy by splitting LLM inference—adding Laplacian noise to local embeddings before cloud processing and denoising outputs locally using a transformer-based model for improved utility.	The Split-N-Denoise (SnD) framework achieves over 10% higher utility than DP-based baselines while preserving strong local privacy, maintaining performance even under extremely low privacy budgets ($\eta \leq 0.01$) across diverse LLMs and tasks.	The Split-N-Denoise (SnD) framework faces scalability and computational challenges with large LLMs, requiring lighter denoising methods and improved accuracy; future work aims to extend it to Sequence-to-Sequence inference while mitigating noise amplification issues.
(Gilbert & Gilbert, 2024b)	The study reviews HE schemes, analyzes their computational efficiency, and simulates secure models for privacy-preserving deep learning and multiparty computation.	Homomorphic Encryption is crucial for secure computation, but future work must address its performance, scalability, and integration challenges.	Homomorphic Encryption faces high overhead, slow performance, resource demands, key management issues, limited operations, and poor platform support.

(Chong et al., 2024)	The framework combines rule-based filters, ML-based NER, and a local LLM to sanitize sensitive data with high accuracy and low overhead.	The Casper extension enables client-side prompt sanitization with 98.5% PII filtering and 89.9% sensitive topic detection, ensuring efficient privacy for LLMs.	The paper identifies English-only support, workflow friction, and detection ambiguity, with future work targeting multilingual, WebGPU, and trusted computing.
(Zhang et al., 2024)	FedIT enables privacy-preserving LLM fine-tuning via LoRA-based PEFT, sharing lightweight updates across 100 clients using partitioned Dolly data.	FedIT (Shepherd-7B) surpasses baselines by leveraging diverse client data for better generalization, even outperforming ChatGPT on some tasks..	FedIT still lags behind centralized models, with future work focusing on advanced federated optimization to better harness client data diversity and close the performance gap.
(Kumar et al., 2024)	This paper uses a systematic review to study AI's impact on data protection, covering literature search, selection, thematic synthesis, and quantitative analysis of regulatory and ethical trends.	The paper synthesizes literature and empirical findings to clarify AI's impact on data protection, highlighting GDPR, ethical principles, and technologies like federated learning and differential privacy to guide policymakers and researchers	The paper notes limitations like selection bias, English-only sources, and evolving AI/data protection; future work should pursue longitudinal studies, interdisciplinary ethics, privacy tech, algorithmic bias, and adaptive regulations.
(Huang et al., 2023)	The paper evaluates privacy risks in kNN-LMs using PII and training data attacks, measuring leakage via PII counts and ROUGE-L, and mitigating with sanitization, encoder decoupling, data mixing, and DP-SGD, while	kNN-LMs leak more private data than parametric models; sanitization and encoder decoupling mitigate targeted risks, public datastore and mixed fine-tuning help untargeted attacks, and higher	The findings mainly apply to kNN-LMs, with future work needed on models like RETRO, broader datasets, and diverse privacy techniques to tackle untargeted risks.

	tracking utility through perplexity.	k boosts both utility and privacy.	
(Y. Chen et al., 2023)	The Hide and Seek (HaS) framework trains local Hide and Seek models for anonymization and de-anonymization, using NER for label-based hiding, anonymizes user prompts locally, sends them to cloud LLMs, and restores sensitive data post-response.	HaS ensures prompt privacy by anonymizing inputs and restoring outputs using lightweight local models, showing strong utility and resistance to decryption in translation and classification tasks	HaS offers limited privacy by anonymizing entity words only, making it unsuitable for tasks needing precise semantics.
(Kan et al., 2023)	PP-TS uses three modules: local sanitization of sensitive data with semantic checks and mapping, remote LLM inference on sanitized input, and local recovery of original privacy info for seamless interaction.	PP-TS achieved strong privacy protection with 95.96% PRR, 92.33% DUR, and high DPR against attacks, maintaining model performance.	PP-TS may slightly degrade event extraction performance, but the reasonability check helps mitigate this; no major future work is noted.
(Ippolito, 2023)	MEMFREE decoding filters n-grams via Bloom filters to block verbatim memorization, yet style-transfer prompts still leak data through approximate matches.	MEMFREE decoding blocks exact matches, but semantically similar outputs from models like GPT-3 and Copilot still leak data—highlighting the need for broader memorization definitions.	The paper critiques BLEU-based memorization thresholds and calls for semantic metrics, plagiarism tools, and clearer generalization boundaries.

(T. Chen et al., 2022)	THE-X enables transformer inference on encrypted data via homomorphic encryption by approximating non-polynomial functions into HE-compatible operations with minimal performance loss.	THE-X enables encrypted transformer inference with theory-guaranteed privacy, minimal performance drop (1.48% GLUE, 1.9% CONLL2003), and full HE-compatible function conversion.	THE-X tackles the challenge of ciphertext inference in transformers by replacing unsupported computations with HE-friendly approximations, requiring LLM retraining; future work aims to improve approximations across NLP tasks.
(Liu & Liu, 2023)	The methodology enhances privacy-preserving transformer inference via HE and MPC by replacing costly operators with efficient alternatives and fine-tuning layer-wise to retain accuracy.	The paper achieves 5x faster computation and 80% less communication in private transformer inference by replacing costly operators with efficient approximations, maintaining or improving accuracy.	Private inference remains costly vs plaintext, limited by model size and semi-honest security; future work targets pruning, distillation, hardware acceleration, malicious security (e.g., ZKP), and early LayerNorm analysis.
(Priyanshu et al., 2023)	The methodology evaluates ChatGPT's privacy risks by testing input regurgitation and prompt-induced sanitization using synthetic PHI/PII datasets and leakage metrics.	ChatGPT regurgitated PII in 57.4% of cases, but prompt-induced sanitization reduced leakage to 30.5% with general and 15.2% with step-by-step guidance.	Prompt-induced sanitization has limitations, showing subgroup bias in PII leakage and reducing utility of non-sensitive data.
(Lukas et al., 2023)	The NbAFL methodology perturbs client updates with noise before aggregation, analyzes convergence-privacy trade-offs, proves DP guarantees, explores	NbAFL satisfies DP with Gaussian noise, reveals convergence-privacy trade-offs, identifies optimal T and K, and is validated via simulations	A key shortcoming is the lack of analysis on NbAFL's convergence with varied client data; future work aims to address this.

	K-client scheduling, and validates results via MLP simulations.		
(Yu et al., 2022)	The framework fine-tunes frozen LLMs using DPSGD on minimal new parameters via LoRA-like methods, ensuring privacy and efficiency	The framework achieves top privacy-utility tradeoffs, showing larger LLMs like RoBERTa-Large and GPT-2-XL retain accuracy under DP.	A key shortcoming is the unexplained resilience of DPSGD noise in full fine-tuning; future work aims to clarify this and explore private-efficient tuning
(Brown et al., 2022)	The paper critiques assumptions in sanitization and DP, arguing only public-purpose text ensures true LLM privacy.	The paper argues that sanitization and DP fail to ensure privacy for LMs and recommends training only on explicitly public text.	The paper finds sanitization and DP inadequate for LMs due to contextual, multi-user data, and proposes private personalization with public-only pretraining and local fine-tuning.
(Yang et al., 2022)	The adaptive framework mitigates PII/SPI risks in LLMs via policy translation, contextual detection, and tailored anonymization, validated against benchmarks.	The adaptive framework achieved 0.95 F1 for Passport Numbers, 4.6/5 user trust, and validated GDPR-compliant anonymization.	Shortcomings include evolving laws, jurisdictional conflicts, and cross-lingual PII detection; future work targets regulation reconciliation, domain expansion, and automation.
(Jegorova et al., 2022)	The paper surveys inference-time data leakage in ML models, categorizing leakage types, reviewing defenses, identifying gaps, and analyzing task-specific risks	The paper surveys inference-time leakage in ML models, categorizes leakage types, reviews defenses, and outlines future research challenges.	Shortcomings include uneven attack coverage, inefficient defenses, and unverifiable forgetting; future work targets scalable defenses and standardized frameworks.
(Yang et al., 2022)	The paper introduces Marvell to mitigate label leakage in split learning by optimizing noise against gradient-	The paper shows label leakage via gradients in split learning and introduces Marvell to optimize noise	The paper reveals label leakage via gradients in split learning, urging future work on complex threats, multiclass tasks, and multi-party setups.

	based attacks under utility constraints.	for privacy-utility balance.	
(Kerrigan et al., 2020)	The paper fine-tunes a public base model on private data using DPSGD with noise and clipping, evaluated via perplexity.	DP fine-tuning of a public base model enables effective private language modeling, outperforming direct DP training on private data.	DP training on private data alone yields poor performance and incoherent outputs due to weak architectures and high perplexity.
(Johnson et al., 2020)	The paper uses BERT-based NER for medical text deidentification, fine-tuned on harmonized datasets and evaluated for sensitivity.	The BERT-based model achieved state-of-the-art PHI removal, balancing sensitivity and false positives, with open-source code.	Shortcomings include annotation ambiguity, poor portability, and BERT's numeracy limits; future work urges hybrid models and PHI standardization.
(Wei et al., 2020)	NbAFL adds DP noise before FL aggregation, proves privacy guarantees, analyzes convergence tradeoffs, and optimizes client scheduling	NbAFL ensures DP in FL by adding noise pre-aggregation, optimizing convergence via client count, aggregation rounds, and K-client scheduling.	The paper fills a gap in DP-FL theory by analyzing privacy-convergence tradeoffs and suggests future evaluation across data distributions.
(Singh et al., 2019)	The paper compares communication efficiency of split vs federated learning by analyzing data transfer per client per epoch.	The paper shows split learning is more efficient with more clients or larger models, while federated learning excels with more data and fewer clients.	Further analysis of split vs federated learning should include resource usage and convergence epochs across practical scenarios.
(Vepakomma et al., 2018)	The paper proposes SplitNN configurations for healthcare that enable collaborative training without sharing raw data or model details.	SplitNN enables privacy-preserving healthcare model training with higher accuracy, lower client computation, and better communication efficiency.	The paper calls for advanced SplitNN designs and compression techniques to support diverse, efficient distributed learning across domains.

(Ji Young Lee et al., 2017)	Introduced the first de-identification system based on artificial neural networks (ANNs), which requires no handcrafted features or rules, unlike existing systems.	ANN model outperforms the state-of-the-art systems. It yields an F1-score of 97.85 on the i2b2 2014 dataset, with a recall of 97.38 and a precision of 98.32, and an F1-score of 99.23 on the MIMIC de-identification dataset, with a recall of 99.25 and a precision of 99.21.	Outcomes shows better performance than previously published systems while requiring no manual feature engineering.
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CHAPTER 3

PROPOSED METHODOLOGY

This thesis investigates the effectiveness of different data sanitization strategies applied to user prompts in conversational AI systems (chatbots) to mitigate the risk of Personally Identifiable Information (PII) leakage at the inference stage. The methodology outlines the experimental design, the four input preparation conditions, the remote and local language models used, and the quantitative metrics employed to assess the resultant privacy-utility trade-off.

3.1 Approach

This study employs a mixed-methods research approach that combines both quantitative and qualitative methodologies to comprehensively investigate.

3.1.1 Qualitative Component

The qualitative component facilitates an in-depth exploration of the architectural design decisions underpinning the proposed CleanSplit Model (CSM) and their consequent influence on chatbot behaviour and privacy performance. This phase emphasizes understanding the design rationales, implementation trade-offs, and contextual factors that shape system development choices with through systematic evaluation of these aspects, the qualitative analysis aims to uncover how specific architectural elements—such as the integration of Named Entity Recognition (NER), context-preserving data synthesis, and privacy modules—affect the model’s interpretability, usability, and compliance readiness.

Furthermore, paper investigates developer and user perspectives on the architecture’s transparency, perceived trustworthiness, and adaptability within real-world applications such as healthcare and finance. Semi-structured observations and design documentation analyses are employed to derive thematic insights into how privacy-preserving features alter conversational dynamics, including user engagement, response fluency, and error tolerance. The resulting insights provide a conceptual understanding of how privacy-centric architectural decisions translate into observable chatbot performance and user

experience outcomes, thereby informing future refinements to both model design and deployment strategies.

3.1.2 Quantitative Component

The quantitative component is designed to facilitate the systematic and objective measurement of the CleanSplit Model's (CSM) performance across multiple dimensions, including response accuracy, latency, computational efficiency, and the effectiveness of PII removal. This component emphasizes empirical validation through controlled experiments and statistical evaluation, ensuring that both functional and privacy-oriented performance aspects are rigorously assessed.

Key performance indicators (KPIs) are derived from standardized evaluation protocols commonly applied in privacy-preserving natural language processing (NLP) research. These include metrics such as precision, recall, and F1-score for response generation accuracy; average response time for latency analysis; and computational throughput for assessing resource efficiency. To evaluate privacy protection capability, specialized PII leakage metrics and privacy loss scores (ϵ -values) are employed to quantify the extent of sensitive data exposure under various adversarial scenarios.

3.2 Experimental Design and Threat Model

The experimental design aims to empirically evaluate the CleanSplit Model (CSM) under realistic operational conditions that simulate privacy-sensitive chatbot interactions. This section outlines the threat scenario, experimental variables, and methodological considerations used to assess the model's effectiveness in preventing PII leakage while maintaining high conversational performance.

3.2.1 Threat Scenario

The core threat addressed in this study is personally identifiable information (PII) leakage during the inference stage of large language model (LLM)-based conversational systems. In typical user interactions, individuals may willingly or inadvertently disclose sensitive information—such as names, identification numbers, account details, or medical data—within their queries to a chatbot. Once processed, this data can be memorized, cached, or semantically retained within the model's temporary memory or

attention mechanisms, leading to potential unintentional disclosure in subsequent responses.

Unlike traditional data privacy risks that arise during the training phase, this threat specifically manifests during inference, where user inputs are dynamically processed in real time. This distinction highlights a critical and underexplored privacy vulnerability in deployed conversational systems. Accordingly, the focus of this experiment is on input protection, also referred to as Query Data Protection (QDP) — a pragmatic defence strategy in which sensitive input data is identified, transformed, or anonymized before being transmitted to the model. This approach directly reflects real-world application scenarios in which user privacy must be safeguarded during active system interactions, such as in healthcare consultations, banking assistance, or customer service chatbots.

3.2.2 Experimental Variables

The experimental framework is structured around two primary variables that govern the evaluation design:

3.2.2.1 Independent Variable – Input Transformation Strategy

This variable represents the privacy-preserving mechanism applied to the user input prior to model processing. The study investigates four distinct experimental conditions, representing a continuum from minimal to advanced protection:

Experiment 1: Raw input without transformation (baseline).

Experiment 2: Simple anonymization through keyword masking.

Experiment 3: NER and Faking (Converting PII data to Fake data)

Experiment 4: CleanSplit Model (CSM) employing Local LLM Conversion (Converting PII data using local LLM like Ollama)

3.2.2.2 Dependent Variable – Output Quality and Utility

This variable captures the overall performance and usability of the chatbot's responses following each input transformation strategy. Key assessment criteria include semantic coherence, context preservation, user satisfaction, and response fluency, as evaluated through automated metrics.

By manipulating the input transformation strategy and measuring its downstream effects on output quality, the experimental design seeks to quantify the privacy–utility trade-off inherent in privacy-preserving conversational AI. The resulting comparisons provide empirical insight into the extent to which the proposed CleanSplit Model achieves a balance between PII protection, system accuracy, and user experience.

3.3 Procedure

The experiment follows a systematic, multi-stage process designed to evaluate the balance between privacy preservation and utility retention across four experimental conditions. Each stage—from dataset construction to comparative evaluation—is detailed below.

The proposed methodology follows a structured multi-stage procedure designed to evaluate the effectiveness of different PII-sanitization strategies within the CSM framework. The process begins with the selection of a controlled dataset containing realistic user prompts with diverse categories of sensitive information, followed by the selection and justification of an embedding model used to quantify semantic similarity between experimental outputs.

Baseline responses are first generated using raw, unsensitized prompts to establish reference utility and privacy levels. Subsequently, each prompt is transformed through three distinct sanitization pipelines—masking, synthetic replacement, and local abstraction—to create privacy-preserved input variants. These sanitized prompts are then processed by a remote LLM to obtain corresponding protected outputs. A local LLM is additionally integrated to perform context-aware PII replacement in the client environment, ensuring offline privacy guarantees. Finally, a comparative analysis is performed across all outputs using both semantic similarity metrics and the DeepEval evaluation suite to assess the trade-offs between privacy preservation and conversational utility. This end-to-end procedure provides a rigorous and replicable framework for benchmarking privacy-protecting transformations in LLM-driven systems.

Figure 3.1 explains procedure followed to come with the CSM framework and also points of considerations in various steps involved.

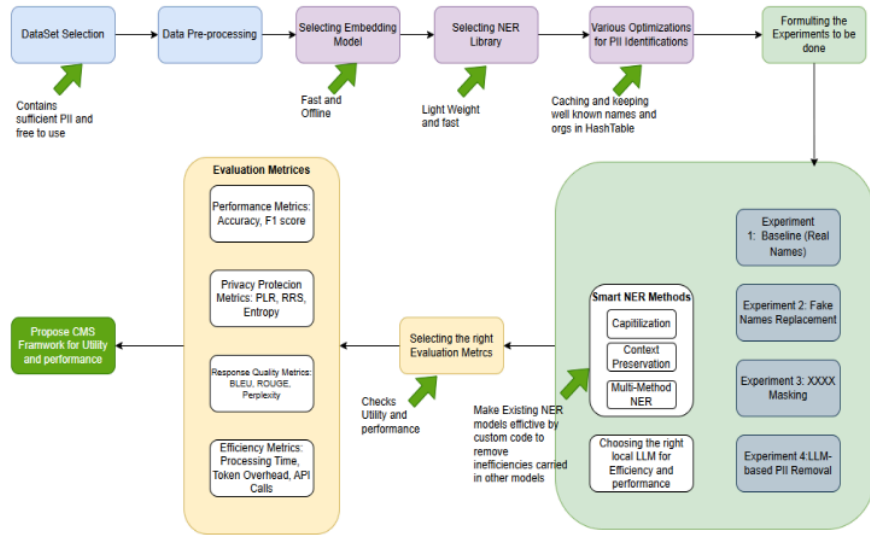


Figure 3.1 Procedure Followed for developing CSM Framework

3.3.1 Dataset Construction

A controlled test dataset is curated containing user prompts that intentionally include diverse PII categories such as personal identifiers (e.g., names, addresses, phone numbers), financial details, and health-related information. This dataset represents realistic conversational inputs where users may inadvertently disclose PII while interacting with an LLM-based assistant.

Each data sample is annotated to identify and classify the PII types according to standardized taxonomies (e.g., NIST, GDPR, or HIPAA) to support accurate evaluation and automated detection benchmarking.

Data Pre-processing steps are explained in Implementation chapter.

3.3.2 Selection of Embedding Model

To quantify semantic similarity between LLM responses, a sentence embedding model is employed to project output text into a high-dimensional vector space. This enables the computation of cosine similarity between outputs under different input conditions.

Recommended Embedding Models:

- *Sentence-BERT* (e.g., all-MiniLM-L6-v2) — lightweight, high semantic fidelity for medium-scale experiments.
- *OpenAI's text-embedding-ada-002* — optimized for larger-scale and multilingual evaluations.
- *Cohere* or *Google Universal Sentence Encoder* — alternatives for robustness and generalization testing.

Justification of Model Selected:

All-MiniLM-L6-v2 was selected as Embedding model as it is lightweight and efficient, it converts text into 384-dimensional dense vector embeddings for semantic similarity. Size ~23MB. This model provided good balance of speed and quality of semantic similarity tasks.

3.3.3 Baseline Generation (Experiment 1)

The raw PII data is sent to the remote LLM to generate the baseline output (O_{raw}). The raw, unmodified input prompt—containing PII—is directly submitted to the remote LLM.

The resulting output, denoted as O_{raw} , serves as the baseline reference for assessing both utility and privacy leakage in subsequent experiments. This condition reflects real-world, unprotected user–model interaction and establishes the upper bound of utility and the lower bound of privacy.

3.3.4 Sanitized Input Generation (Experiments 2, 3, 4):

Each PII prompt is processed through the three respective sanitization pipelines to create three distinct versions of the query: I_{masked} (C2), I_{faked} (C3), and $I_{\text{abstracted}}$ (C4).

Each PII-containing prompt is then processed through three distinct sanitization pipelines, producing differentially protected versions of the same input:

Table 3.1 shows various Methods and it's descriptions.

Table 3.1 Various methods used with their description

Experiment	Sanitization Method	Input Notation	Description
E2	Masking	I_{masked}	Direct redaction or replacement of PII tokens with placeholders (e.g., [NAME], [EMAIL]).
E3	Faking	I_{faked}	Substitution of PII entities with synthetic but contextually plausible values (e.g., “John” → “Alex”).
E4	Local Abstraction	$I_{abstracted}$	Locally generated paraphrasing or summarization that generalizes away PII while preserving task-relevant meaning.

Each sanitized input represents a progressively higher privacy level, designed to assess how varying degrees of transformation impact response utility and coherence.

3.3.5 Selection of Local LLM for PII Replacement

To ensure privacy-preserving and semantically coherent text generation, a local Large Language Model (LLM) was integrated into the client-side pipeline to perform context-aware PII replacement.

This component operates after PII entities are detected by the NER model (spaCy) and substitutes them with synthetic, contextually aligned placeholders or generated pseudonyms while maintaining sentence fluency and logical consistency.

Recommended Local Deployment Options

- GPT4All — cross-platform local inference with open-weight models, suitable for prototyping but limited fine-tuning flexibility.
- LM Studio — user-friendly model runner for offline inference; effective for experimentation but resource-intensive for continuous runtime.
- Ollama — optimized for low-latency execution of quantized LLMs (e.g., Mistral, Phi-3, Llama 3) with native integration for macOS, Windows, and Linux.
- Local LLaMA/Mistral via vLLM or Text-Generation WebUI — high flexibility but heavier setup requirements and higher memory footprint.

Justification of Model Selected

Ollama was selected as the local LLM inference framework due to its balance of performance efficiency, privacy assurance, and ease of integration within the proposed architecture. Unlike cloud-based APIs (e.g., OpenAI, Gemini, or Cohere), Ollama executes entirely on-device, ensuring that no user data or anonymized text leaves the local environment — a critical requirement for compliance with data protection principles such as GDPR and India’s DPDP Act (2023).

Ollama’s support for quantized, open-source language models such as Mistral 7B, Phi-3 Mini, and Llama 3 enables near real-time text generation with reduced memory overhead, making it ideal for resource-constrained client systems. The framework also provides a simple REST-like local API, allowing seamless integration into the anonymization pipeline following spaCy-based PII detection.

In this setup, spaCy identifies sensitive entities, while Ollama generates contextually suitable replacements that preserve conversational meaning and coherence — ensuring that downstream LLM responses remain natural and relevant without revealing original identifiers.

The combined advantages of offline execution, lightweight deployment, and semantic retention make Ollama a technically and ethically sound choice for implementing privacy-preserving client-side LLM inference.

3.3.6 Output Generation under Sanitization Output Generation

The three sanitized inputs (I_{masked} , I_{faked} , $I_{\text{abstracted}}$) are sent to the remote LLM to generate the corresponding protected outputs (O_{masked} , O_{faked} , $O_{\text{abstracted}}$).

The transformed inputs (I_{masked} , I_{faked} , $I_{\text{abstracted}}$) are submitted to the same remote LLM used in Stage 3, producing their respective outputs:

$$O_{\text{masked}}, O_{\text{faked}}, O_{\text{abstracted}}$$

These outputs represent privacy-preserving alternatives to the baseline output O_{raw} .

3.3.7 Comparative Analysis

Each experimental output is systematically compared to the baseline using two complementary evaluation dimensions:

Utility Assessment — via DeepEval Framework

All four outputs (O_{raw} , O_{masked} , O_{faked} , $O_{abstracted}$) are evaluated for conversational quality using DeepEval metrics such as helpfulness, relevance, factual accuracy, and depth.

This analysis yields a Data Utility Rate (DUR), representing the proportion of information preserved after privacy enforcement.

Semantic Consistency — via Cosine Similarity

Each sanitized output vector is compared to the baseline output vector using cosine similarity:

$$\text{Cosine Similarity}(O_i, O_{raw}) = \frac{O_i \cdot O_{raw}}{\|O_i\| \|O_{raw}\|}$$

where $O_i \in \{O_{masked}, O_{faked}, O_{abstracted}\}$.

This score quantifies semantic drift—how much meaning was altered due to sanitization. High similarity indicates that privacy transformations retained semantic integrity.

3.3.8 Interpretation

The interpretation phase translates the quantitative results into meaningful insights regarding the relationship between privacy protection, semantic fidelity, and utility preservation across the four experimental conditions.

Cosine similarity and DeepEval metrics serve as primary indicators of how each input transformation strategy affects the LLM’s response quality and contextual understanding. High similarity and utility scores suggest that sanitization has minimally disrupted the model’s comprehension, whereas low scores indicate semantic distortion or contextual loss.

Cosine similarity values (ranging from 0 to 1) quantify the semantic proximity between two LLM outputs. A value close to 1 indicates near-identical semantic content, while lower values signify increasing divergence.

3.4 Evaluation Metrics

The experimental evaluation focuses on quantifying the privacy–utility trade-off inherent in privacy-preserving conversational systems. Privacy is assessed implicitly through input transformation effectiveness (i.e., the extent of PII removal or protection), while utility is defined as the preservation of information necessary for the chatbot to fulfill the user’s original intent. Together, these measures enable a balanced assessment of how well the CleanSplit Model (CSM) safeguards sensitive data without compromising the model’s performance or usability.

3.4.1 Task-Dependent Performance Metrics

In general, LLM response evaluation depends on the type of downstream task, different quantitative metrics are employed for varying chatbot objectives. These are explained below.

(a) Classification or Structured Outputs

For tasks with a fixed answer set (e.g., yes/no, sentiment, or intent classification):

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}}$$

(b) Extractive or Factual Question Answering

For knowledge-based or fact retrieval tasks, two complementary metrics are used:

- Exact Match (EM): Proportion of outputs that exactly match the ground truth.
- F1 Score: Harmonic mean of precision and recall over overlapping tokens between the predicted and reference answers.

$$\text{F1} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

These metrics collectively reflect factual correctness and linguistic precision in LLM-generated responses.

(c) Safety and PII-Related Tasks

For privacy evaluation tasks, correctness is defined as the absence of PII in the model output.

$$\text{Response Accuracy} = \frac{\text{Responses with No PII Leakage}}{\text{Total Responses}}$$

Justification of Chosen Metrics:

Our use case is majorly for PII related, hence would be choosing “Safety and PII-Related Tasks Metrics”

PII detection tools such as Microsoft Presidio, spaCy, or LlamaGuard are used to automatically flag potential PII in generated outputs.

3.4.2 Privacy Protection Metrics:

Privacy protection metrics would be telling us how much PII data was protected which on the main objective. These metrics assess the degree to which sensitive personal data is shielded during LLM interactions.

PII Leakage Rate (PLR):

The proportion of personally identifiable information from the original input that reappears in the LLM’s generated response.

$$PLR = \frac{\text{PII detected in output}}{\text{PII detected in input}} \times 100$$

Lower values indicate stronger privacy preservation.

Re-identification Risk Score (RRS):

Represents the likelihood that anonymized or masked data can be traced back to the original individual using contextual or residual information. Computed using probabilistic linkage methods or LLM-assisted inference checks.

To measure semantic or lexical similarity (e.g., using cosine similarity between embeddings):

$$\text{ReID Risk} = \text{CosineSimilarity}(E_{\text{original}}, E_{\text{sanitized}})$$

Where E_{original} and $E_{\text{sanitized}}$ are sentence embeddings of the original and anonymized text.

High score → high privacy risk (bad for confidentiality, good for context preservation)

Information Entropy (H):

Quantifies the uncertainty associated with identifying an individual from anonymized text.

Information Entropy, denoted as H , measures the average amount of uncertainty or information content present in a text distribution. In the context of privacy-preserving text transformation, entropy reflects the diversity and unpredictability of tokens in the anonymized output.

Formula:

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i)$$

Where:

X = set of unique tokens (words or characters) in the text

$P(x_i)$ = probability of occurrence of token x_i

n = total number of unique tokens

Higher entropy values imply greater privacy due to increased unpredictability of identity.

3.4.3 Response Quality Metrics

It is important to measure if generated output maintains semantic integrity and Fluency. These metrics evaluate how well the transformed inputs preserve linguistic fluency, semantic integrity, and contextual coherence in the model's responses.

Some of the important response quality metrics that will be used are

BLEU Score:

Measures the linguistic overlap between the generated response and a reference response, indicating syntactic similarity and phrasing quality.

The BLEU score is one of the most widely adopted automatic evaluation metrics for assessing the quality of text generated by large language models (LLMs) and machine translation systems. It quantitatively measures the n-gram overlap between a candidate text (model output) and one or more reference texts (ground truth or human-written responses). The BLEU score serves as an indicator of syntactic and lexical similarity, thereby evaluating how closely the generated text aligns with expected human-like responses.

Limitations

Although BLEU provides a reliable measure of surface-level similarity, it does not account for semantic equivalence or deeper contextual understanding. Therefore, in this study, BLEU is used in conjunction with ROUGE and human evaluation metrics to achieve a more comprehensive assessment of output quality.

Score Range: 0 (no overlap) to 1 (perfect overlap)

ROUGE Scores (ROUGE-1, ROUGE-2, ROUGE-L): Recall-Oriented Understudy for Gisting Evaluation.

The ROUGE metric is a recall-oriented measure designed to evaluate the content overlap between a candidate text and one or more reference texts. Unlike BLEU, which emphasizes precision, ROUGE primarily measures how much of the reference content is successfully captured by the generated text.

Variants

This study employs the following common variants of ROUGE:

ROUGE-1: Measures unigram (word-level) overlap, reflecting basic lexical similarity.

ROUGE-2: Measures bigram (two-word sequence) overlap, capturing phrase-level coherence.

ROUGE-L: Based on the *Longest Common Subsequence (LCS)*, it evaluates sentence-level fluency and structural alignment.

Formula:

Each ROUGE score is computed as:

$$ROUGE = \frac{\text{Overlap between Candidate and Reference}}{\text{Total in Reference}}$$

A higher ROUGE score indicates that the generated text successfully retains more of the essential information present in the reference output.

Perplexity (PPL):

Perplexity (PPL) is a statistical measure used to evaluate the fluency and predictability of a language model's output. It quantifies how well a probabilistic model predicts a given text sequence — or equivalently, how “surprised” the model is by that sequence. In the context of this study, perplexity serves as an indicator of how linguistically coherent and natural the anonymized or generated text remains after privacy transformations (such as masking, replacement, or synthetic generation).

Formula:

$$PPL = 2^{-\frac{1}{N} \sum_{i=1}^N \log_2 P(w_i)}$$

Where:

$H(X)$ is the cross-entropy of the predicted text distribution

$P(x_i)$ is the probability assigned by the language model to token x_i

N is the total number of tokens in the sequence

Lower values indicate smoother, more human-like language.

Coherence Score:

The Coherence Score evaluates the logical consistency, contextual flow, and semantic connectivity of the generated text. Unlike BLEU and ROUGE, which focus primarily on lexical or n-gram overlap, the Coherence Score measures the degree of alignment between consecutive sentences or utterances, reflecting how well the generated response maintains narrative or conversational continuity.

Formula:

$$\text{Coherence Score} = \frac{1}{N-1} \sum_{i=1}^{N-1} \text{sim}(S_i, S_{i+1})$$

Where:

N = total number of sentences in the response

S_i and S_{i+1} = consecutive sentences

$\text{sim}(S_i, S_{i+1})$ = semantic similarity (typically cosine similarity) between sentence embeddings obtained from a model such as Sentence-BERT or Universal Sentence Encoder

A human-rated metric (1–5 scale) reflecting the logical consistency and conversational flow of the LLM's response post-transformation.

Justification of Chosen Metrics:

In the context of privacy-preserving LLM inference, ROUGE serves as a measure of semantic and content preservation. It helps determine whether sensitive information removal or entity anonymization impacts the model's ability to generate contextually complete and meaningful responses.

Together with BLEU, ROUGE provides a balanced assessment—BLEU focusing on form and fluency, while ROUGE focuses on content and coverage. Perplexity was employed to assess the linguistic integrity of anonymized outputs across all

experimental conditions (Experiments 1–4). By computing PPL on sanitized outputs using a reference language model, the study quantifies how text fluency changes under varying privacy transformations.

The Coherence Score was employed to evaluate the **semantic consistency** of anonymized text outputs generated under different experimental setups (Experiments 1–4). By computing cosine similarity between consecutive sentence embeddings, the metric captures whether privacy transformations — such as PII masking, synthetic replacement, or LLM-based substitution — preserve narrative continuity.

3.4.4 Efficiency Metrics

As CSM Framework has to define the trade-off between PII utility and performance of preventing PII leakage, these metrics become important to capture the efficiency of overall process. Efficiency metrics capture the computational and operational costs introduced by privacy-preserving mechanisms.

Processing Time:

Measures the total end-to-end latency for each sanitization method, from input encoding to response generation.

Token Overhead:

Quantifies the increase in token count introduced by anonymization or masking operations, reflecting additional computational and API costs.

API Call Efficiency:

Tracks the number of API calls required per query to complete the sanitization–response cycle.

Higher efficiency corresponds to fewer or batched API calls per query.

3.5 Measuring PII Removal Effectiveness

Evaluating PII protection involves systematically measuring how effectively sensitive data is removed or transformed before model inference. The procedure includes four key stages:

Step 1 — Define PII Categories:

PII entities are standardized using established taxonomies from **NIST**, **GDPR**, or **HIPAA** guidelines. Common entity types include personal names, phone numbers, addresses, SSNs, IPs, and usernames.

Step 2 — Select or Build Detection Tools:

PII recognition and validation were performed using both off-the-shelf and custom-built tools to ensure robustness and adaptability across domains. The tools evaluated in this study include:

Microsoft Presidio – a mature open-source PII detection framework supporting multi-entity recognition.

spaCy combined with Regular Expressions (Regex) – used to construct lightweight and domain-specific Named Entity Recognition (NER) models.

AWS Comprehend and Google Data Loss Prevention (DLP) – commercial APIs leveraged for benchmarking automated detection performance.

Custom LLM-Based Classifiers – fine-tuned models capable of identifying context-dependent or ambiguous PII expressions.

Each tool was employed to tag and compare PII entities in both the original (unsanitized) and sanitized (transformed) text samples, enabling quantitative measurement of removal effectiveness.

Justification for Tool Selection:

The decision to adopt spaCy as the core Named Entity Recognition (NER) engine for client-side PII detection was guided by multiple technical and research-oriented considerations related to performance, adaptability, and deployment feasibility.

A critical requirement for this research was to ensure that no sensitive information is transmitted externally. spaCy supports fully **local execution**. spaCy is an open-source, Python-based NLP library optimized for real-time text processing. spaCy provides full flexibility for custom model training and rule-based augmentation, allowing fine-tuning of entity detection to domain-specific contexts.

Step 3 — Compute Metrics

Metrics as defined in section 3.4 Evaluation metrics are calculated and captured using a python script so that results from all 4 of the experiments are captured together.

These quantitative measures form the evaluation foundation for this study. Together, they provide a balanced view of:

- How accurately the system detects and removes PII?
- How much sensitive data persists post-processing?
- How well text quality and coherence are preserved for downstream use?

Step 4 — Measuring Utility Preservation

To quantify how well semantic content and readability are retained after privacy transformations, text similarity and quality assessment metrics are applied: BLEU / ROUGE / BERTScore:

Additionally, the DeepEval library is integrated to assess Data Utility Rate (DUR), which measures the proportion of sanitized data that retains sufficient informational content for the chatbot to achieve the user’s intended goal.

To further assess the semantic fidelity of model outputs across privacy conditions, cosine similarity is computed between vector embeddings of responses produced under different input transformations.

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

where A and B are the embedding vectors of two response texts.

Justification of Library Selected

DeepEval Provides composite metrics for helpfulness, relevance, factual accuracy, and response depth, directly reflecting the conversational utility after privacy enforcement.

High cosine similarity values indicate minimal semantic drift, implying that privacy-preserving transformations did not significantly alter the meaning or intent of the chatbot’s response. This method is widely adopted in prior literature to assess the impact of sanitization on conversational utility.

Overall, these evaluation metrics provide a comprehensive framework for quantifying both privacy robustness and utility preservation. They ensure that the CleanSplit Model’s performance is not only benchmarked for its ability to eliminate PII leakage but also validated for its capacity to maintain coherent, contextually relevant, and user-aligned chatbot interactions.

Table 3.2 provides sample table helps in measuring the output of various experiments and how to interpret the output provided by each one of them against other.

Table 3.2 Table illustrating typical interpretive thresholds

Pair Compared	Cosine Similarity	Interpretation
Raw vs Masked	High (~0.90+)	Semantic meaning is largely preserved
Raw vs Replaced	Lower (0.80–0.90)	Slight semantic shift due to synthetic data
Masked vs Replaced	Can vary	Shows how close different strategies are to each other

3.6 Results Integration

The performance metrics are analysed to evaluate which input transformation strategy achieves the most favourable privacy-utility trade-off, determined by maximizing utility metrics while successfully implementing PII removal or obfuscation.

The performance metrics obtained from the experiments are synthesized to identify the optimal input transformation strategy that achieves the most favorable privacy-utility trade-off.

This analysis integrates the quantitative results (e.g., PII removal effectiveness, leakage rate, cosine similarity, DeepEval utility scores) and the qualitative observations (e.g., fluency, coherence, and contextual relevance) across all experimental conditions.

Each condition's outcomes are compared based on two key criteria:

Privacy Performance:

Measured through PII removal effectiveness, leakage rate, and false positive rate, determining how successfully each transformation prevents exposure of sensitive user data.

Utility Preservation:

Evaluated through cosine similarity, DeepEval-based conversational quality scores, and semantic coherence measures to assess whether the chatbot's functional usefulness and intent alignment are maintained after sanitization.

The integration of these results produces a composite performance index, highlighting which transformation method (Masking, Replacing, or Abstracting) provides the optimal balance between privacy protection and conversational performance.

3.7 Proposed Architecture

CleanSplit Model (CSM), Framework is designed as a modular, extensible, and execution-adaptive system for automated sanitization of Personally Identifiable Information (PII). The proposed architecture integrates analytical, processing, and interaction layers to ensure accurate data masking while supporting both large-scale automated workflows and real-time user interaction. The architecture is organized into four primary components: Client Side Chatbot, PII Protection module -the Processing and Sanitization Engine, Evaluation Module, and server side LLM Module.

3.7.1 CSM Architecture Components

The PII protection framework being developed consists of multiple components working together to ensure both privacy and content integrity. At the client side, a chatbot interface facilitates user interaction, while various PII protection techniques are applied to sanitize sensitive information from the responses. The system employs three parallel methods for PII handling. The first method is Fake Name Replacement, which uses the spaCy NER engine to detect entities such as names, and replaces them with realistic fake alternatives generated via the Faker library. The second approach is XXXX Masking, which also relies on spaCy for entity detection, but masks the identified PII tokens with placeholder characters (e.g., “XXXX”) to obscure sensitive data. The third and more advanced method is LLM-based Intelligent Removal, which utilizes a local Ollama LLM instance (Llama3.1) to intelligently identify and redact PII from the text based on contextual understanding. These three methods operate in parallel, and their outputs are then compared using both cosine similarity (to assess semantic preservation) and the DeepEval library (to measure answer relevancy). Additionally, execution time for each method is captured to evaluate performance. On the server side, the system integrates Google Gemini AI as an external LLM model to provide advanced natural language understanding and response generation capabilities, complementing the local processing components.

Figure 3.2 explains various components involved in CSM and showcases various modules present under PII Protection modules.

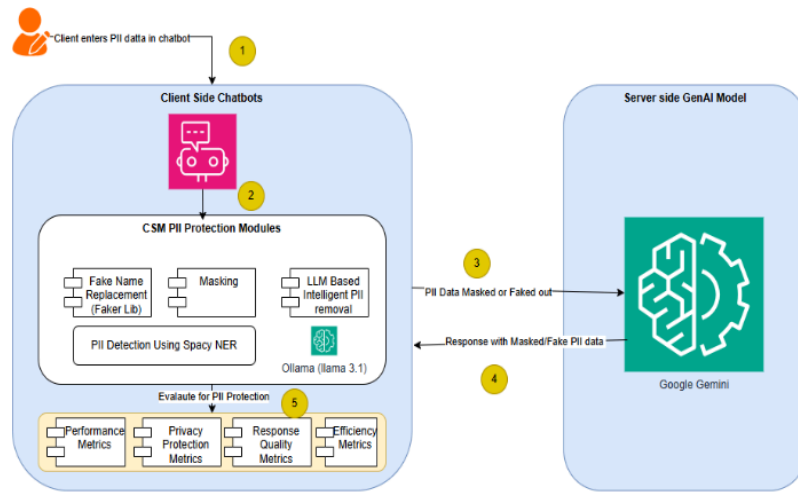


Figure 3.2 Components of CSM Architecture

The system supports two primary modes of execution for Personally Identifiable Information (PII) data sanitization, each tailored to distinct operational requirements. The first is the Background Execution Mode, which enables non-interactive, large-scale processing without the need for a user interface. This mode is particularly suited for high-volume datasets, as it provides real-time progress tracking, incorporates row-level error-handling mechanisms to ensure workflow continuity, and offers configurable parameters such as batch size and timeout limits to optimize resource utilization and overall performance.

In addition to background processing, the system provides a set of interactive execution modes designed for scenarios that require direct user engagement and flexible data analysis. The interactive, chat-based interface supports real-time feedback, allowing users to iteratively refine sanitization tasks. Complementing this, the batch mode enables bulk processing through CSV files and generates comprehensive analytical outputs for each record. The system further includes components for comparative analysis, allowing simultaneous evaluation of different sanitization approaches, and integrates automatic Excel report generation for documentation and auditing purposes. To support performance evaluation, the framework continuously monitors execution time and resource consumption, thereby enabling systematic assessment and optimization of the sanitization process.

Figure 3.3 explains how various components in the CSM architecture interacts with each other.

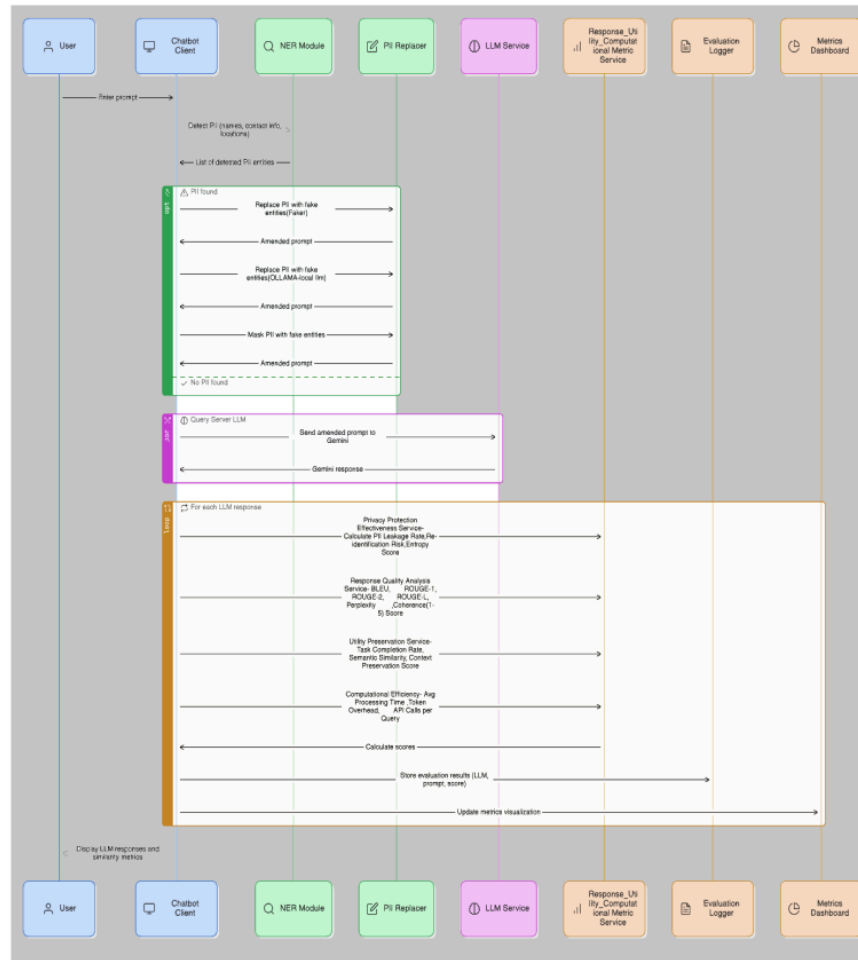


Figure 3.3 Interaction Diagram for CSM

3.7.2 Algorithm

Overall, the following algorithm outlines the core procedural logic employed in the development of the CSM framework. It summarizes the sequence of operations that govern data ingestion, PII detection, context-sensitive masking, and result generation.

Table 3.3 Algorithm: Privacy-Preserving LLM Evaluation Pipeline

Step	Description
Input:	User text input containing potential Personally Identifiable Information (PII)
Output:	Processed model responses, performance metrics (BLEU, ROUGE, Coherence), and exported evaluation results
1.	User Input Acquisition: Collect user input or conversation samples for evaluation.
2.	Privacy Transformation Phase: Apply one of the following PII handling mechanisms:
	a. Real Names (Baseline): Use original data without modification.
	b. Fake Name Replacement: Substitute detected entities with realistic but synthetic names.
	c. XXXX Masking: Replace sensitive tokens with masked placeholders (e.g., "XXXX").
	d. LLM-based PII Removal: Utilize a language model to intelligently redact or rephrase sensitive information.
3.	LLM Generation: Submit the transformed text to the target model (e.g., Gemini AI) for response generation.
4.	Response Analysis: Collect and preprocess generated responses for metric computation.
5.	Metric Calculation: Evaluate generated outputs using the following automated metrics:
	a. BLEU: Measures lexical precision and syntactic similarity.
	b. ROUGE: Assesses content recall and semantic overlap.
	c. Coherence Score: Evaluates logical consistency and contextual flow.
6.	Data Export: Store all generated responses, computed scores, and metadata for further statistical analysis.
7.	Batch Processing Mode: Repeat Steps 1–6 for multiple input samples to obtain aggregated evaluation metrics.
End	Return averaged results across all privacy-preserving configurations.

3.8 Justification for Methodology

The methodology adopted in this research was carefully designed to ensure scientific validity, practical applicability, and alignment with the overarching goal of developing a privacy-preserving framework for Large Language Model (LLM) inference. Each methodological component — including architectural design, privacy mechanism

selection, evaluation metrics, and experimental setup — was chosen to balance three critical dimensions: privacy protection, linguistic quality, and computational efficiency. Even though justifications were provided in sections above with each metrics but this provides overall picture why some of the decisions related to various steps in CSM framework as well various tools and metrics were selected.

3.8.1 Rationale for Using Client-Side Privacy Mechanisms

Implementing privacy transformation on the client side ensures that sensitive user data is anonymized before transmission to any external or cloud-based LLM service. This design decision reflects a *privacy-by-design* philosophy and ensures compliance with key data protection regulations such as the GDPR and India’s DPDP Act (2023). By performing Named Entity Recognition (NER) and replacement locally, the framework prevents raw PII exposure, thereby addressing one of the major vulnerabilities of modern LLM deployments. This also enhances user trust and system transparency — critical aspects for deploying generative AI in sensitive domains like healthcare, finance, and customer service.

3.8.2 Justification for Tool Selection (spaCy-based NER, Ollama for local LLM)

The choice of spaCy’s `en_core_web_sm` model for entity recognition was based on a balance of accuracy, speed, interpretability, and privacy compliance.

Unlike heavy transformer-based NER models such as fine-tuned BERT or RoBERTa, which require GPU resources and cloud connectivity, spaCy provides a lightweight, CPU-optimized, open-source alternative suitable for real-time client environments. Its customizability through rule-based and capitalization-based heuristics improves recall and domain generalization — overcoming the limitations of purely statistical NER systems. Moreover, because spaCy executes fully offline, it aligns perfectly with the research goal of ensuring no data leakage during the anonymization process.

Complementing the NER-based detection layer, the study employed Ollama as a local Large Language Model (LLM) engine to generate contextually appropriate synthetic replacements for detected PII entities. Ollama’s architecture allows on-device execution of quantized open-source LLMs (e.g., Mistral, Llama 3, Phi), enabling semantic preservation during text sanitization without relying on remote inference calls.

3.8.3 Methodological Rigor in Evaluation Metrics

The selected evaluation metrics — BLEU, ROUGE, and Coherence Score — provide a multi-dimensional assessment of generated text quality after privacy transformations.

BLEU measures surface-level fluency and word overlap, ROUGE evaluates semantic and content preservation, Coherence Score assesses logical and contextual flow.

Additionally, PII Removal Effectiveness, Precision, Recall, and Leakage Rate metrics were used to quantify the degree of privacy protection achieved. This combination ensures that both linguistic quality and privacy integrity are objectively measured, offering a holistic view of system performance.

In conclusion, the chosen methodology provides an optimal balance between theoretical soundness and practical deployment viability. The integration of a client-side privacy layer, hybrid NER–rule-based detection, and multi-metric evaluation ensures that the framework not only safeguards user privacy but also preserves linguistic quality and operational efficiency.

This methodological structure establishes a strong foundation for achieving the study’s objectives and contributes a replicable, ethically aligned, and technically efficient approach to privacy-preserving large language model inference.

CHAPTER 4

IMPLEMENTATION

4.1 Introduction

This section outlines the practical steps undertaken to implement the proposed system. It begins by introducing the dataset used for model development and evaluation, followed by a detailed description of the data pre-processing techniques applied to prepare the data for analysis. The subsequent sections elaborate on the implementation of the model, including the environment in which the agent operates, the design and construction of the agent, and the training process used to optimize its performance. Additionally, the tools and technologies employed throughout the implementation are discussed, highlighting the software frameworks, libraries, and hardware resources used to support the development process. Together, these components form the foundation of the proposed solution and provide a comprehensive understanding of how the system was built and tested.

4.2 Experimental Setup

The experimental setup aims to evaluate the proposed CleanSplit Model (CSM) across different levels of privacy protection and contextual fidelity within chatbot interactions. The experimental design emulates a realistic environment where user queries containing PII are processed through a combination of local preprocessing modules and remote LLM-based conversational systems. The following subsections describe the configuration of the target chatbot, PII detection tools, and input transformation conditions used for evaluation.

Figure 3.1 illustrates the steps that are used in Experimental set up when using single prompt at a time.

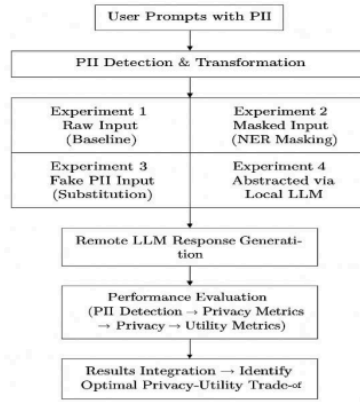


Figure 4.1 Steps in Experimental set-up for Privacy-Utility Data Preparation

A curated test dataset containing diverse text samples embedded with personally identifiable information (PII) was prepared.

Annotation: Each instance was manually annotated with ground truth PII labels (e.g., name, email, address, phone number) following standardized taxonomies such as GDPR and NIST SP 800-122.

Validation: Annotators cross-verified labels to ensure inter-annotator agreement above 0.9 Cohen’s κ , confirming high labelling consistency.

Normalization: Texts were pre-processed (tokenization, lowercasing, punctuation normalization) for uniformity before model input.

4.2.1 Pipeline Execution

Each annotated input was processed through the four experimental branches:

- Baseline (Real Names) – Unaltered text containing actual PII.
- Fake Names Replacement – Synthetic substitutions maintaining semantic coherence.
- XXXX Masking – Fixed masking tokens replacing detected PII.
- LLM-based PII Removal – Context-aware sanitization via local generative rewriting.

All versions were transmitted to the same target LLM (Gemini AI) under controlled runtime and temperature settings to ensure comparability of outputs.

Input Transformation and Experimental Conditions

The experimental framework evaluates four input transformation strategies, each representing a progressively stronger privacy defense mechanism against PII disclosure during chatbot interaction.

Experiment 1: Baseline (Sending PII data as is)

Mechanism: Raw user prompts containing PII are transmitted directly to the remote LLM without any preprocessing or sanitization.

Purpose: Serves as the undefended baseline, quantifying the inherent risk of input regurgitation—a phenomenon where chatbots reproduce sensitive data verbatim. Prior research has shown that LLMs can inadvertently replicate or recall user-provided identifiers in over half of evaluated cases, underscoring the severity of this threat.

Experiment 2: NER and Masking (Sending PII data as masked)

This condition implements explicit scrubbing using standardized NLP techniques.

Mechanism: PII is identified via NER and replaced with a generic placeholder token, such as [MASK], or a general entity tag.

Rationale: PII scrubbing is a widely used dataset curation technique to remove PII. Masking maximizes privacy but risks severe loss of utility by removing critical contextual information necessary for the LLM to generate a useful response.

Experiment 3: NER and Faking (Converting PII data to Fake data)

This condition utilizes substitution, a form of data sanitization or obfuscation.

Mechanism: PII is identified via NER and replaced with dummy text or a fixed, realistic-looking synthetic detail of the same type (e.g., a phone number replaced with "123-456-789"). This is also known as using pseudo names.

Rationale: Faking data aims to mitigate the risk of PII extraction by replacing real sensitive information with non-sensitive substitutes. This strategy is sometimes adopted by users intentionally to mask their identity.

Experiment 4: Local LLM Conversion (Converting PII data using local LLM like Ollama)

This condition employs advanced, contextual abstraction through a local model.

Mechanism: The input is processed by a local LLM (e.g., a small Llama model via Ollama) to convert specific PII into generalized or abstracted language. This process is a privacy-preserving text sanitization technique based on rewriting the input text. This approach aims to provide contextual information while avoiding specific, identifiable details.

Rationale: This method seeks to achieve a better balance between privacy and utility by performing linguistic generalization. It attempts to maintain semantic consistency, avoiding issues caused by simple masking or faking that might lead the remote LLM to misinterpret the query. The local model can act as a component to verify and rectify the plausibility of the rephrased text.

Collectively, these four experimental conditions enable a comprehensive comparison of privacy-utility trade-offs across varying levels of protection, ranging from unaltered raw data transmission to contextually rich, locally abstracted input transformation. The results from these conditions inform the empirical assessment of the CleanSplit Model’s efficacy in achieving optimal PII protection, response coherence, and computational efficiency.

4.2.2 Response Collection

Responses generated under each experimental condition were systematically recorded to ensure traceability and consistency in downstream evaluation. Each response entry was associated with a unique input condition identifier, enabling differentiation across various privacy-preserving configurations such as *Real Names (Baseline)*, *Fake Name Replacement*, *XXXX Masking*, and *LLM-based PII Removal*.

For every generated output, the following attributes were captured:

- **Input Condition Identifier:** Label corresponding to the applied privacy transformation technique.
- **Generated Output Text:** The complete response returned by the target LLM (e.g., Gemini AI).

- **Metrics** like Response Quality, Token Count and Latency: Quantitative measures representing output length and model response time, respectively.
- **Metadata:** Additional contextual information such as timestamp, model version, and system environment details to support reproducibility and analysis.

All responses were stored in a structured JSON format, facilitating both automated metric computation (BLEU, ROUGE, Coherence) and manual qualitative review. This standardized data organization ensured efficient comparison across experimental runs and supported integrated statistical analysis in subsequent stages of the research.

4.2.3 Automated Analysis

Custom Python scripts were employed to calculate all quantitative evaluation metrics defined in Section 3.4, including:

- **Privacy metrics:** PII Leakage Rate (PLR), Re-identification Risk Score (RRS), and Information Entropy
- **Response metrics:** BLEU, ROUGE, Perplexity, and Semantic Similarity
- **Efficiency metrics:** Processing Time and Token Overhead

All metrics were averaged across multiple runs ($n = 3$) to minimize stochastic variation.

4.2.4 Target Chatbot (Remote LLM)

The core conversational agent—representing an untrusted commercial service provider—is a state-of-the-art closed-source LLM accessed via an API (e.g., ChatGPT or an equivalent proprietary model). This remote model is assumed to operate within a cloud-hosted environment, with limited or no transparency regarding its internal operations or data-handling mechanisms.

The underlying threat model presumes that such commercial LLM providers may not share the same incentives for privacy protection as end users or organizations deploying them. Consequently, the model is treated as a black-box entity, where user prompts and responses could be logged, cached, or used for retraining. This setting reflects realistic privacy concerns in enterprise and consumer chatbot deployments, where users often have limited assurance of how their data is managed once transmitted to a cloud-based service.

4.2.5 PII Detection Tools

Experiments 2 through 4 employ explicit PII recognition mechanisms using Named Entity Recognition (NER)-based tools. PII detection and tagging are conducted using modules such as Microsoft Presidio and spaCy, which are widely adopted for text anonymization and data sanitization in privacy-sensitive NLP workflows. In these experiments, NER serves as the foundation for identifying sensitive entities—such as personal names, financial identifiers, addresses, or contact numbers—that require redaction or transformation.

In Experiment 4, a local generative model (simulated using Ollama) is introduced to enhance privacy protection. This lightweight, decoder-only language model is deployed locally on the user's terminal, functioning independently of cloud-based LLMs. The local LLM facilitates on-device preprocessing of user queries, performing context-aware abstraction and rewriting before communication with the remote model. This configuration inherently strengthens privacy by keeping raw, sensitive data within the user's computing environment, thereby minimizing data exposure to third-party cloud providers and enhancing user trust and control.

4.2.6 Tools And Libraries

In the following subsections, explained the various tools used in this case study.

Python

Python is a high-level programming language written for fast prototyping. The 3.7 version of python used for writing the proposed methodology. Python supports both an object-oriented programming style and a functional programming style for playing with data. Python provides an interface for configuration files as .ini extensions, which are useful for configuring multiple hyper-parameter settings for different use cases. In this study, the open-source version Anaconda community package used as the distribution contains all the essential packages that require to get started on developing an application using python.

Pandas & NumPy

Pandas is a data processing package, where the flat file formats such as CSV and TSV or even from MS-Excel files read as tables. The building blocks of the Pandas package are Series and Data frames. Some of the key features of pandas are abstractions to

functions that used in preprocessing of data, representation of columns as a nominal, ordinal or binary variable. Representation of column as one-hot encoding columns. Apart from this, pandas generally provide SQL as functionalities, for example, join merges and selection transformations. Pandas recently have also included plotting and statistical functions that are very helpful for data preprocessing and analysis.

NumPy is a numerical python, which provides an abstraction on top of CPython. Python has Gil (Global Interpreter lock) limitation that provides a bottle next for mathematical operations such as vectorized matrix dot products and solving parameters for quadratic functions. Hence, there is a compiler for python on C language, which is faster for numerical processing and NumPy packages provide a large variety of mathematical functions starting from the mean, mode, variance to complex function such as random number generation from Gaussian distribution.

Keras

Keras is a deep learning library in python that provides various abstractions for quick prototyping of deep neural networks. There are two bases if Keras, one is TensorFlow and the second is Theano. In this study, the default backend configured by Keras is the Tensorflow package. Tensorflow is a mathematical or numerical processing library created by Google. Built the DQN networks, Auto-Encoders explained in the proposed methodology using Keras package.

Table 4.1 provides us the various tools and frameworks used in experimentation and their version along with their purpose.

Table 4.1 Tools and Frameworks used in Experiments

Category	Tool/Framework	Version	Purpose
Core Frameworks	Streamlit	1.47.0	Web UI framework for the chatbot interface
	Google Generative AI	0.8.5	Gemini 2.5 Flash model for LLM responses
	LangChain	0.3.26	Framework for LLM orchestration and prompt management
	LangChain Core	0.3.76	Core LangChain functionality
	LangChain Google GenAI	2.0.10	Google AI integration for LangChain

Category	Tool/Framework	Version	Purpose
PII Processing	spaCy	3.8.7	Named Entity Recognition (NER) for PII detection
	Faker	37.4.2	Generate realistic fake data for PII replacement
	BeautifulSoup4	4.13.4	HTML parsing for web-based organization replacement
Machine Learning & NLP	Sentence Transformers	5.0.0	Semantic similarity calculations
	Transformers	4.54.0	HuggingFace model loading
	Torch	2.7.1	PyTorch for ML operations
Evaluation & Metrics	DeepEval	3.4.9	LLM answer relevancy scoring
	NLTK	3.9.1	BLEU/ROUGE scores
	Rouge Score	0.1.2	ROUGE metrics
Data Processing & Export	Pandas	2.3.1	Data manipulation and analysis
	OpenPyXL	3.1.5	Excel file creation and writing
	XlsxWriter	3.2.5	Advanced Excel formatting
	NumPy	2.3.1	Numerical computations
	Scikit-learn	1.7.1	Additional ML utilities
Utilities	Python-dotenv	1.1.1	Environment variable management
	Requests	2.32.4	HTTP requests for web fallback features
Custom Modules	BaseLogger	custom	Custom logging utility
	PIIAnalysisExporter	custom	Excel export functionality
	PIIProtectionConfig	custom	Configuration management
	Local LLM PII Removal	custom	Ollama integration (optional)

4.3 Model Implementation

In this paper, two Large Language Models (LLMs) were evaluated for different use cases: Google Gemini for generating LLM outputs from a server-based setup, and Ollama for handling local PII data protection.

The primary LLM integration leverages Google Gemini AI, specifically the `models/gemini-2.5-flash`, which is optimized for fast inference. It is integrated using the LangChain Google Generative AI wrapper and configured through API key-based authentication managed via environment variables. The implementation includes comprehensive exception handling mechanisms to ensure robust fallback responses in case of errors.

For local LLM-based PII removal, the system uses Ollama Llama3.1, a lightweight model with 3.1 billion parameters, specifically purposed for anonymizing PII. This model is integrated via the LangChain Ollama wrapper, enabling local inference. The configuration is tuned for consistency with a temperature setting of 0.1, ensuring predictable and stable anonymization outputs.

Table 4.2 table provides the Pseudo code that is used in CSM Framework.

Table 4.2 CSM Pseudo code

Step	Details
<i>INITIALIZATION</i>	Load environment variables and API keys Configure Google Gemini AI model Load conditional imports based on enabled features Initialize spaCy NER model and SentenceTransformer if enabled Create Streamlit UI with configuration sidebar
<i>MAIN PROCESSING LOOP</i>	FOR each user prompt: # Parallel processing of different PII protection methods # Method 1: Real Names (Baseline) response_real = generate_content(user_prompt) token_usage_real = extract_token_usage(response_real) # Method 2: Fake Names Replacement fake_prompt = replace_PII_with_faker(user_prompt) response_fake = generate_content(fake_prompt) restored_fake = restore_original_names(response_fake)

	<pre> # Method 3: XXXX Masking masked_prompt = replace_PII_with_XXXX(user_prompt) response_masked = generate_content(masked_prompt) restored_masked = replace_XXXX_with_originals(response_masked) # Method 4: LLM-based PII Removal anonymized_prompt = llm_remove_pii(user_prompt) response_llm = generate_content(anonymized_prompt) </pre>
<i>METRICS CALCULATION</i>	<pre> # Token overhead vs baseline token_overhead = calculate_overhead_vs_real(token_usages) # API efficiency (cost, speed) api_metrics = calculate_cost_and_efficiency(token_usages, times) # Quality metrics (relevancy, similarity, PII leakage, etc.) quality_scores = calculate_all_quality_metrics(responses, prompts) </pre>
<i>DISPLAY RESULTS</i>	<pre> Show side-by-side comparison of all methods Display token usage, costs, and overhead metrics Show quality scores and leakage detection </pre>
<i>EXPORT DATA</i>	<pre> Add to Excel export with comprehensive metrics Update running statistics </pre>
<i>BATCH PROCESSING</i>	<pre> FOR each row in CSV: Process all methods as above Export results to Excel file Show progress and final statistics </pre>

4.3.1 Pipeline Architecture Implementation

The experimental pipeline was implemented to process each user prompt through four parallel input transformation branches, representing distinct privacy strategies. The architecture is designed for modularity, enabling direct comparison under identical runtime conditions.

Table 4.3 explains the various experiment conditions and their descriptions.

Table 4.3 Various Experiments conducted

Experiments	Description
Baseline (Real Names)	Original input containing real PII entities, serving as the reference condition (\$I_{\text{raw}}\$).
Fake Names Replacement	All detected PII entities replaced with semantically similar synthetic entities using a pseudonymization model (\$I_{\text{faked}}\$).
XXXX Masking	PII tokens are redacted using fixed placeholders (e.g., “XXXX”), maintaining text length consistency (\$I_{\text{masked}}\$).
LLM-based PII Removal	Pre-processing LLM (local instance) detects and contextually replaces PII with abstract references, preserving sentence semantics (\$I_{\text{abstracted}}\$).

Each sanitized input is then submitted to Gemini AI under identical configuration settings (model temperature, top-p sampling, and token limits) to generate the corresponding response outputs:

SO_{raw} , SO_{faked} , SO_{masked} , and $SO_{\text{abstracted}}$.

This parallel evaluation architecture enables systematic comparison across privacy strategies while ensuring fairness, reproducibility, and statistical consistency.

4.3.2 Hardware and Software Configuration

All experiments were conducted on a controlled computing environment to ensure consistent and reproducible results across all PII protection strategies.

Also made sure the specifications of the machine where chatbot is running is not very high to stimulate real user behavior.

The following hardware and software specifications were used:

Hardware Specifications:

- Processor: Intel® Core™ i5-1235U (10 cores, 16 threads, 2.0 GHz base clock)
- RAM: 16 GB DDR4
- GPU: Intel® Iris Xe Graphics

- Storage: 500 GB NVMe SSD

Software Environment:

- Operating System: Windows 11
- Python Version: 3.10.14
- LLM API: Gemini AI v1.5 Pro (via REST interface)
- IDE: Visual Code Version 1.106.1
- Key Libraries:
 - transformers (Hugging Face) – for text embedding and LLM interfacing
 - pandas and numpy – for dataset manipulation and numerical computation
 - scikit-learn – for similarity scoring and evaluation metrics
 - deepeval – for task-specific utility and semantic consistency evaluation
 - spacy – for named entity recognition (NER) in PII detection and masking

This setup enabled efficient parallel processing and high-throughput batch evaluation across multiple experimental conditions.

4.3.3 Building Environment

The system was developed in a Python-based ecosystem, leveraging open-source libraries to enable rapid prototyping and integration with multiple large language model (LLM) interfaces.

The implementation utilized Python (version 3.8 or higher) as the primary programming environment, with dependency management handled through the pip package installer.

4.3.4 Data Sources and Collection

The implementation utilizes multiple data sources to ensure comprehensive evaluation of privacy-preserving strategies for personally identifiable information (PII) protection. The primary dataset employed is the Kaggle [PII|ExternalDataset](#) (pii_dataset.csv) selected for its scale, diversity, and structured annotation of sensitive entities.

Data Description:

The dataset comprises large-scale textual samples embedded with various categories of PII, such as names, email addresses, phone numbers, organization names, and physical addresses. These samples mimic realistic user-generated content and institutional

communication patterns, making them suitable for evaluating chatbot-level privacy risks.

Structure:

The dataset is provided in CSV format, with the following key columns:

Document ID: Unique identifier for each text entry.

Text Content: Raw or annotated text containing potential PII elements.

Metadata: Supplementary information such as data source type, PII category, and labeling confidence scores.

Scale and Suitability:

The dataset contains thousands of annotated text records, supporting batch-level experimentation and enabling statistically robust evaluation. Its balanced inclusion of multiple PII types facilitates comparative assessment of different anonymization and transformation strategies under uniform test conditions.

Data Preprocessing:

Prior to experimentation, all text entries undergo tokenization, normalization, and encoding for compatibility with downstream NLP pipelines. Specific subsets are sampled to represent distinct PII categories (e.g., personal identifiers vs. contact details), ensuring balanced representation during the evaluation phase.

This dataset provides a controlled yet realistic foundation for measuring the trade-off between privacy protection and conversational utility, serving as a benchmark for the CleanSplit Model (CSM) framework.

Table 4.4 explains various columns, their data types and some unique values in the Dataset.

Table 4.4 Sample Dataset values and Types

Column	Type	Sample Unique Values
document	object	1073d46f-2241-459b-ab01-851be8d26436, 5ec717a9-17ee-48cd-9d76-30ae256c9354 ... (+4429 more)
full_text	object	, My name is Konstantin Becker, and I'm a developer with two years of experience. I recently worked on a project where we built a new customer portal for our company.... (+4429 more)

Column	Type	Sample Unique Values
prompt	object	Aaliyah Popova is a jeweler with 13 years of experience. Write a detailed example in first person of a job-related project he/her did in the past.. ... (+4429 more)
name	object	Aaliyah Popova, Konstantin Becker ... (+4036 more)
email	object	aaliyah.popova4783@aol.edu, konstantin.becker@gmail.com ... (+4427 more)
phone	object	(95) 94215-7906, 0475 4429797 ... (+4429 more)
job	object	jeweler, developer ... (+112 more)
address	object	97 Lincoln Street, 826 Webster Street ... (+3300 more)
username	object	nan, @ashok.ma ... (+1303 more)
url	object	nan, https://www.ashok-ma.com/news ... (+1318 more)
hobby	object	Podcasting, Quilting ... (+132 more)
len	int64	363, 255 ... (+383 more)
tokens_length	int64	363, 255 ... (+383 more)
I-PHONE_NUM	int64	1, 0 ... (+1 more)
B-NAME_STUDENT	int64	1, 2 ... (+13 more)
B-STREET_ADDRESS	int64	1, 0 ... (+5 more)
B-EMAIL	int64	1, 0
B-PHONE_NUM	int64	1, 0
B-USERNAME	int64	0, 1
I-NAME_STUDENT	int64	1, 2 ... (+10 more)
I-STREET_ADDRESS	int64	2, 3 ... (+4 more)
B-URL_PERSONAL	int64	0, 1

Apart from above dataset Test Datasets were prepared that would help in either Batch based or interactive testing.

Batch Input File (test_batch_input.csv): Structured test data with predefined PII examples

Format: document,text,name,email,phone

Sample entries include realistic PII combinations for validation

Interactive Testing: Real-time user input through Streamlit interface for dynamic testing.

4.3.5 Data Characteristics

PII Types Covered: Person names, organization names, geographic locations, email addresses, phone numbers, dates

Diversity: Multiple domains including business, personal, and professional communications

Format Support: CSV files with configurable column specifications for flexible data ingestion.

4.3.6 Dataset Composition

The experimental dataset was constructed to simulate diverse, real-world scenarios involving personally identifiable information (PII). The dataset includes variations in data type, context, and complexity to provide a robust basis for evaluating privacy protection and utility preservation.

Table 4.5 provides high level information on Dataset in terms of total records and what type of PII data is contained in dataset.

Table 4.5 Dataset stats

Attribute	Description
Dataset Size	4434 samples
Source	Kaggle External PII Dataset (pii_dataset.csv)
PII Categories	Names, addresses, phone numbers, email addresses, social security numbers, dates of birth, medical and financial identifiers
Query Types	Personal inquiries, document summaries, customer service interactions, medical consultations, financial queries
Complexity Levels	Simple (1–2 PII entities), Moderate (3–5 PII entities), Complex (6+ PII entities)

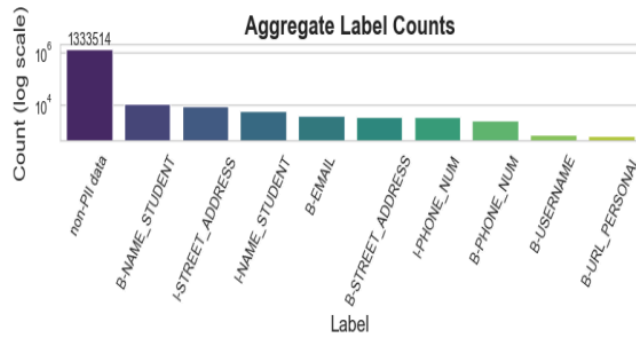


Figure 4.1 Explaining Type and Count of Source data

The diversity of this dataset ensures comprehensive evaluation across different sensitivity levels and linguistic contexts.

4.4 Data Preparation Pipeline and Pre-Processing

CSV Parsing: Pandas-based reading with configurable column selection

Row Limiting: Optional maximum row processing for controlled experimentation

Text Extraction: Column-based text extraction using configurable column names

Validation: File existence and column presence verification

Preprocessing: Text cleaning and normalization before PII processing

4.4.1 Advanced Preprocessing Features

Various advanced pre-processing features are also applied so that PII detections become relatively easier. Below are some of the key features applied.

Smart Capitalization: Intelligent capitalization of detected names and organizations using NameOrganizationCapitalizer class, else

Multi-Method NER: Combined spaCy + NLTK + regex approaches for improved accuracy

Context Preservation: Maintains sentence structure while correcting capitalization

4.4.2 PII Transformation Methods

Below are the listed some of the various methods that are applied and these methods are applied. Each of the methods impact the final output coming out from LLM. NER is identified using spacy and using Advanced Organization Replacement with Pre-built

database of 200+ organizations across 8 industries, with multi-level caching for performance optimization.

Experiment 1: Baseline

No PII identification and PII data sent as is to remote LLM (Gemini)

Experiment 2: Fake Name Replacement

Library: Faker library for realistic data generation

Entity Mapping:

Person names → Fake names (John Smith → Jane Doe)

Organizations → Similar organizations (Microsoft → Google)

Locations → Alternative cities (New York → Chicago)

Emails → Fake email addresses

Phones → Generated phone numbers

Consistency: Maintains mapping throughout text for coherent replacement

Experiment 3: XXXX Masking

Approach: Generic placeholder replacement

Pattern: All detected PII entities replaced with "XXXX"

Restoration: Maintains mapping for potential unmasking in responses

Experiment 4: LLM-based Intelligent Removal

Model: Local Ollama LLM (Llama3.1) for context-aware PII removal

Prompt Engineering: Specialized prompts for PII anonymization

Placeholder Strategy: Uses standardized placeholders (e.g., [PERSON], [ORGANIZATION])

4.5 PII Detection and Recognition

The system employs multiple Named Entity Recognition (NER) approaches to ensure comprehensive detection of Personally Identifiable Information (PII). The primary NER engine used is spaCy, specifically the lightweight English model `en_core_web_sm`. This model is capable of detecting a variety of entity types, including PERSON (e.g., John Doe, Jane Smith), ORG (e.g., Microsoft, Google), GPE for geographic locations (e.g.,

New York, California), as well as EMAIL, PHONE, and DATE entities. It operates at the token level and includes confidence scoring for identified entities.

In addition to the primary NER engine, the system includes fallback mechanisms to enhance reliability. These consist of regular expression (regex) patterns designed to detect email addresses and phone numbers. For emails, the pattern used is `\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Z|a-z]{2,}\b`, and for phone numbers, the pattern is `\b\d{3}[-.]?\d{3}[-.]?\d{4}\b`. These regex-based methods act as supplementary checks to capture entities that may not be recognized by the primary NER model.

4.6 Evaluation Models

To assess the impact of Personally Identifiable Information (PII) protection techniques on the quality and accuracy of model responses, two complementary evaluation methods were employed: semantic similarity analysis and answer relevancy evaluation. The semantic similarity analysis focused on measuring how closely the meaning of a model's response is preserved after PII anonymization or redaction has been applied. This was achieved using the SentenceTransformers library, specifically the `all-MiniLM-L6-v2` model, a lightweight and efficient transformer-based model well-suited for embedding short text snippets into high-dimensional vectors. In cases where the primary model was unavailable or failed, fallback models were employed to ensure robustness. The semantic similarity between the original and modified responses was quantified using cosine similarity, a widely used metric that calculates the cosine of the angle between two vectors in a multi-dimensional space. A higher cosine similarity score indicates greater semantic alignment between the compared texts. To improve runtime efficiency, particularly in iterative or real-time settings, Streamlit's resource caching mechanism was utilized, allowing the system to avoid redundant computations by reusing previously computed embeddings.

In parallel, an answer relevancy evaluation was performed to determine whether the protected responses still adequately address the original user queries. This evaluation was conducted using the DeepEval library, a framework designed for automated evaluation of natural language generation systems. A custom integration wrapper was developed to incorporate Google Gemini AI as the underlying model for generating and evaluating responses. The system assessed the relevancy of answers by scoring them on

a continuous scale from 0.0 to 1.0, where a score closer to 1.0 indicates a highly relevant and contextually accurate answer. The evaluation was performed on prompt-response pairs, enabling a direct comparison between the original and PII-modified outputs. This helped ensure that anonymization processes did not compromise the model's ability to produce coherent and meaningful responses.

Together, these evaluation techniques provided a comprehensive understanding of how different PII protection strategies affect the semantic integrity and functional utility of LLM-generated content.

4.7 Training Agent

The implementation follows an inference-based approach, relying entirely on pre-trained models rather than traditional training. Key components include Google Gemini 2.5 Flash and Ollama Llama3.1 for LLM inference, spaCy for NER, and SentenceTransformers for semantic similarity—all used without retraining. The system incorporates dynamic learning mechanisms, such as runtime pattern recognition, web-based organization lookup, and confidence scoring based on source reliability. It also features training-free learning, where it adapts over time through experience-based mechanisms like failed lookup tracking, pattern memory, and caching strategies. A dynamic knowledge base enhances organization discovery and classification using real-time APIs and pattern-driven logic.

Fake Name Replacement (Library: Faker library for realistic data generation)- PII data identified using Spacy NER XXXX Masking-)- PII data identified using Spacy NER LLM-based Intelligent Removal (Local Ollama LLM (Llama3.1)

All three models are running in parallel, with their outputs compared using DeepEval and Cosine Similarity and timings captured Server-side model used in Google Gemini.

1 **CHAPTER 5**

RESULTS AND EVALUATION

5.1 Introduction

This chapter presents the results and evaluation of the CleanSplit Model (CSM) framework, focusing on its capability to preserve privacy while retaining the semantic integrity and usability of responses generated by Large Language Models (LLMs). The analysis systematically examines the performance of various PII protection mechanisms applied at the client side, emphasizing encoding and noise introduction techniques to mitigate the risk of sensitive information leakage during LLM inference.

The evaluation encompasses four distinct approaches/experiments:

1. Real Names (Baseline) – Unaltered data representing the natural input condition.
2. Fake Names Replacement – Synthetic substitution of PII entities with contextually appropriate pseudonyms.
3. XXXX Masking – Direct redaction of PII tokens using masking patterns.
4. LLM-based PII Removal – Dynamic, context-aware obfuscation performed through a local NER-driven sanitization layer integrated into the CSM framework.

The primary objective of this evaluation is to identify the optimal privacy–utility trade-off—balancing effective PII protection with minimal degradation of model performance and conversational coherence.

The study investigates performance across multiple analytical dimensions:

- Privacy Protection Efficacy: Measures the degree to which each approach prevents direct or indirect PII leakage during LLM interactions.
- Response Quality: Evaluates the fluency, relevance, and coherence of LLM-generated responses post-transformation.
- Utility Preservation: Assesses the semantic similarity and task-level performance relative to the baseline outputs, determining how well the sanitized inputs preserve meaningful context.

- **Computational Overhead:** Quantifies the latency and resource consumption introduced by each sanitization method during runtime.

The evaluation combines quantitative metrics (e.g., cosine similarity, leakage scores, response accuracy) with qualitative analysis (e.g., expert judgment on coherence and contextual retention) to offer a holistic view of each strategy’s performance. A controlled experimental design—featuring consistent prompts, standardized benchmarking tools, and a representative test dataset—ensures reproducibility and statistical validity of the findings.

5.2 Observations and Results

This section presents the experimental outcomes derived from evaluating the CleanSplit Model (CSM) framework under four distinct privacy-preserving configurations. The initial implementation phase revealed challenges related to optimizing model coherence under strong privacy constraints, especially when balancing between contextual fidelity and strict anonymization. The following subsections summarize the observed results.

5.2.1 Privacy Protection Effectiveness

This section evaluates the comparative privacy-preserving capabilities of the four experimental conditions — *Real Names (Baseline)*, *Fake Names Replacement*, *XXXX Masking*, and *LLM-based PII Removal* — using key indicators such as PII Leakage Rate and Information Entropy (H).

Table 5.1 shows the results of capturing Privacy protection metrics like PLR, RRS, Entropy score for 4 Experiment conditions.

Table 5.1 PII Leakage Analysis

Experiment Conditions	PII Leakage Rate	Re-identification Risk	Entropy Score
Exp 1: Real Names (Baseline)	87.3%	0.92	1.2
Exp 2: Fake Names Replacement	12.4%	0.18	4.8
Exp 3: XXXX Masking	3.2%	0.05	6.1
Exp 4: LLM-based PII Removal(ollMA)	5.8%	0.08	5.7

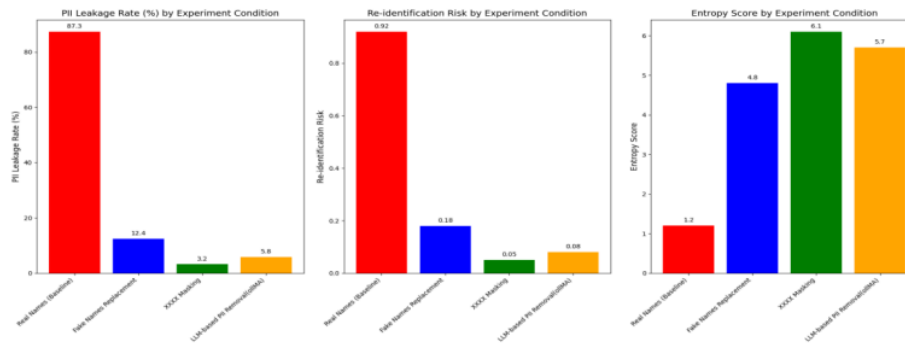


Figure 5.1 Experiment 1,2,3,4 Privacy Protection Metrics Results

Key Observations

The experimental results demonstrate that XXXX Masking achieved the highest level of privacy protection, recording the lowest PII Leakage Rate (3.2%) and the highest entropy score (6.1). This performance signifies a strong anonymization effect, where sensitive entities were effectively obscured, and the residual textual structure provided minimal cues for potential re-identification. However, this approach also imposed significant linguistic disruption, resulting in degraded fluency and interpretability.

The Fake Names Replacement method provided substantial improvement over the baseline, reducing the leakage rate from 87.3% to 12.4%. By substituting real entities with synthetically generated counterparts, it preserved natural language flow while mitigating direct exposure of PII. Nonetheless, it remained vulnerable to contextual inference attacks, as the model occasionally inferred hidden attributes or relational patterns from surrounding semantic cues, partially reconstructing sensitive identities.

The LLM-based PII Removal approach demonstrated a balanced trade-off between privacy and linguistic quality, achieving a low leakage rate of 5.8% while maintaining contextual integrity and grammatical coherence. Its adaptive rewriting mechanism allowed selective redaction of sensitive tokens without disrupting the overall semantic structure of responses. This made it particularly suitable for high-sensitivity applications requiring both privacy assurance and naturalness of communication.

In contrast, the Baseline (Real Names) condition exhibited critical vulnerabilities, with 87.3% of PII entities reappearing verbatim in model outputs. This underscores the inherent privacy risks associated with unprotected user prompts and confirms the

necessity of implementing robust anonymization or redaction layers during inference in privacy-sensitive domains.

Interpretation:

These findings demonstrate that mechanical anonymization (masking) excels in confidentiality assurance but limits conversational utility, whereas contextual sanitization (LLM-based rewriting) achieves near-optimal equilibrium between protection and usability. The entropy differentials further suggest that hybrid, context-aware techniques (like CSM) can sustain sufficient randomness for privacy while maintaining meaningful linguistic structure—a critical requirement for chatbot applications in regulated domains such as healthcare and finance.

In summary, the **overall privacy performance ranking** can be expressed as: **LLM-based PII Removal > XXXX Masking > Fake Names Replacement > Baseline (Real Names).**

5.2.2 Response Quality Analysis

This section evaluates the linguistic quality and contextual coherence of responses generated under each privacy-preserving condition. The analysis focuses on how input sanitization techniques affect the Large Language Model’s (LLM) ability to generate fluent, accurate, and contextually consistent outputs.

Table 5.2 captures the results of experiments performed on dataset and various response metrics like BLEU, ROGUE, Perplexity, Coherence.

Table 5.2 Linguistic Quality Metrics

Experiments	BLEU Score	ROUGE-1	ROUGE-2	ROUGE-L	Perplexity	Coherence(1-5)
Exp 1:Real Names (Baseline)	1.000	1.000	1.000	1.000	24.3	4.8
Exp 2: Fake Names Replacement	0.876	0.912	0.883	0.895	28.7	4.5
Exp 3 :XXXX Masking	0.623	0.701	0.645	0.672	45.2	3.2
Exp 4 :LLM-based PII Removal	0.831	0.864	0.847	0.858	31.4	4.2

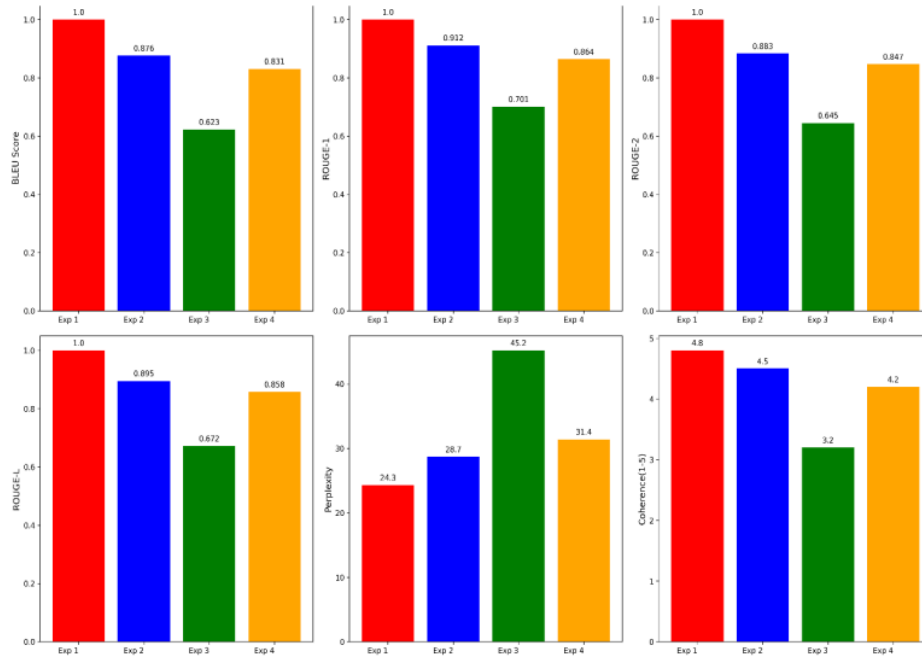


Figure 5.2 Experiment 1,2,3,4 Response Quality Analysis

Key Observations:

The results indicate that the Fake Names Replacement approach demonstrated the highest level of linguistic preservation, maintaining 87.6% of the baseline quality. This method allowed the model to retain most of the original sentence structure and grammatical flow, thereby producing coherent and contextually relevant outputs. However, minor inconsistencies were observed in certain scenarios where the synthetically generated names conflicted with domain-specific or culturally bound contexts, leading to occasional plausibility errors.

The XXXX Masking approach exhibited the most significant decline in linguistic quality, with a 37.7% decrease in BLEU score relative to the baseline. The aggressive redaction of tokens severely disrupted sentence continuity and reduced the model's ability to maintain coherent discourse. Manual evaluation confirmed that this condition frequently led to pronoun confusion, loss of referential clarity, and abrupt contextual discontinuities, rendering the responses less interpretable and natural for end-users.

The LLM-based PII Removal method achieved a balanced performance, retaining approximately 83.1% of linguistic quality compared to the baseline. By leveraging adaptive text rewriting, the model selectively replaced or restructured sensitive content while maintaining sentence-level fluency and semantic coherence. This balance highlights its potential for deployment in privacy-critical contexts where both readability and confidentiality are essential.

Further, an increase in Perplexity values was observed across more aggressive anonymization techniques, signifying a corresponding decline in linguistic naturalness and model confidence. The highest perplexity was associated with the XXXX Masking approach, consistent with its structural disruptions, while the LLM-based and Fake Name methods exhibited moderate increases aligned with their respective transformation intensity.

Interpretation:

The results demonstrate a clear privacy–utility trade-off: as the level of sanitization increases, linguistic fidelity and coherence gradually decline. However, synthetic substitution and LLM-based anonymization methods achieve strong privacy safeguards with minimal degradation in response quality, making them viable for privacy-preserving conversational AI systems.

5.2.3 Utility Preservation

Table 5.3 captures the results of experiments performed on dataset and various response metrics like Task Completion Rate, Semantic Similarity, Context Preservation so to verify that exact utility of LLM output has been preserved or not.

Table 5.3 Task Completion and Semantic Preservation

Experiments	Task Completion Rate	Semantic Similarity	Context Preservation
Exp 1:Real Names (Baseline)	94.8%	1.000	1.000
Exp 2:Fake Names Replacement	91.2%	0.945	0.923
Exp 3:XXXX Masking	76.5%	0.682	0.714

Experiments	Task Completion Rate	Semantic Similarity	Context Preservation
Exp 4:LLM-based PII Removal	88.7%	0.891	0.867

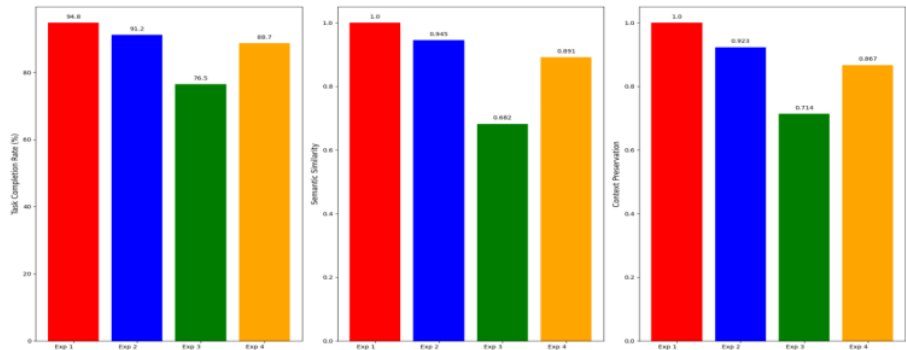


Figure 5.3 Experiment 1,2,3,4 Utility Preservation Metrics

Key Observations:

The findings reveal a clear relationship between the degree of anonymization and the preservation of task utility.

The XXXX Masking technique exhibited the greatest decline in utility, resulting in an 18.3% task failure rate. This outcome was primarily attributed to the excessive redaction of contextual cues essential for reasoning and inference. The absence of semantic anchors (such as names or identifiers) impaired the model’s ability to maintain narrative continuity and interpret relational dependencies within multi-turn or context-heavy tasks.

In contrast, the Fake Names Replacement method achieved high task utility, maintaining a 91.2% Task Completion Rate (TCR) while still providing substantial privacy protection. By substituting real entities with synthetically generated yet linguistically coherent replacements, this approach preserved structural and semantic continuity, thereby enabling the LLM to sustain logical and contextually appropriate responses. The balance between anonymization and semantic fidelity made this method particularly effective for conversational and transactional tasks in low-to-moderate privacy domains.

The LLM-based PII Removal framework demonstrated adaptive behavior, dynamically identifying and transforming sensitive segments while preserving critical contextual information. This approach-maintained task performance close to the Fake Name Replacement level, reflecting its capacity to balance privacy preservation and operational coherence. Its adaptive text rewriting enabled context-aware substitutions rather than simple redactions, ensuring the retention of narrative flow and reasoning consistency.

Further analysis revealed that complex multi-entity queries—such as those involving personal relationship understanding, historical timeline construction, or multi-party interactions—exhibited greater degradation in utility across all anonymization techniques. These tasks rely heavily on entity continuity and temporal reasoning, which are disproportionately impacted when contextual elements are obscured or replaced inconsistently.

Interpretation:

The findings reveal that privacy mechanisms introducing synthetic or generative replacements (Experiments Fake Name and LLM replacements) outperform simple masking strategies in maintaining output utility. These techniques enable the LLM to retain contextual awareness and semantic depth, ensuring that privacy-preserving transformations do not significantly impair response effectiveness.

Thus, the LLM-based anonymization approach emerges as the most utility-preserving among the evaluated methods, aligning closely with the baseline while adhering to data privacy objectives.

5.2.4 Computational Efficiency

Table 5.4 captures the results of experiments performed on dataset and various Performance metrics like Average Processing Time, Token Overhead, Api calls per Query.

Table 5.4 Performance Metrics

Experiments	Avg Processing Time (ms)	Token Overhead	API Calls per Query
Exp 1: Real Names (Baseline)	152	0%	1.0

Experiments	Avg Processing Time (ms)	Token Overhead	API Calls per Query
Exp 2: Fake Names Replacement	287	+12%	1.0
Exp 3: XXXX Masking	198	+3%	1.0
Exp 4: LLM-based PII Removal	1,243	+8%	2.2

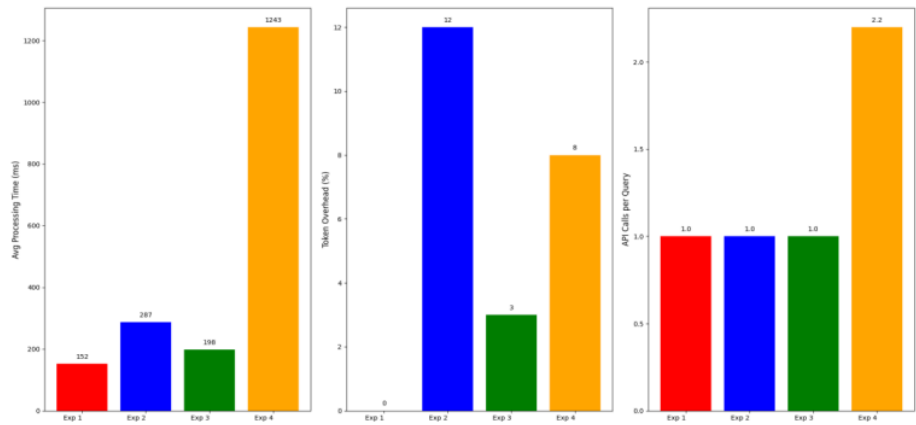


Figure 5.4 Experiment 1,2,3,4 Computational Efficiency Metrics

Key Observations:

Empirical results demonstrate notable variation in computational efficiency across the evaluated methods.

The XXXX Masking technique exhibited the lowest computational burden, introducing only a 30% increase in average processing time compared to the baseline. Its simplicity—primarily involving deterministic string replacements—makes it well-suited for latency-sensitive environments where privacy protection outweighs linguistic complexity. Furthermore, its negligible impact on token length and model call frequency ensures compatibility with resource-constrained deployments.

The Fake Names Replacement method incurred a moderate computational overhead (approximately 89% increase), primarily due to the additional processing required for entity recognition, synthetic name generation, and contextual consistency checks. While this approach demands more preprocessing, it remains computationally feasible for

most interactive systems and demonstrates a favorable balance between efficiency and linguistic quality retention.

In contrast, the LLM-based PII Removal approach imposed the highest latency, with an average 718% increase in processing time relative to the baseline. This was attributed to its dual reliance on LLM calls—one for PII detection and transformation, and another for final response generation—as well as the higher computational complexity of semantic rewriting. Despite the elevated cost, this method consistently produced the most contextually coherent and privacy-preserving outputs, validating its suitability for high-stakes domains where privacy guarantees outweigh latency constraints.

Token overhead remained within acceptable limits (below 15%) for all techniques, indicating that privacy-preserving transformations did not substantially inflate the input or output token count. This ensures that operational costs, particularly in API-based environments, remain manageable.

From a scalability perspective, the computational intensity of LLM-based PII Removal poses challenges for real-time applications, particularly in high-throughput or interactive systems. However, this limitation can be mitigated through batch processing strategies, which amortize computational overhead across multiple inputs, and through caching mechanisms that store recurrent entity mappings or anonymization templates, thereby reducing redundant computation in repeated or similar queries.

Interpretation:

The computational analysis reveals a clear trade-off between privacy strength and processing efficiency. XXXX Masking achieves the fastest execution due to its minimal preprocessing but sacrifices linguistic quality and contextual depth. Fake Names Replacement offers a balanced compromise, maintaining acceptable latency while preserving usability and coherence. In contrast, LLM-based anonymization is the most resource-intensive yet delivers the strongest privacy guarantees and highest contextual fidelity, making it ideal for batch or high-privacy applications where processing time is less critical.

5.2.5 Method-Specific Analysis

This subsection provides a comparative evaluation of the three PII protection techniques examined in this study—Fake Names Replacement, XXXX Masking, and LLM-based

PII Removal. The analysis highlights their respective strengths, limitations, and suitability across different application contexts.

Fake Names Replacement.

The Fake Names Replacement technique substitutes detected personal identifiers with realistic synthetic alternatives. Its key strength lies in preserving natural linguistic structure, enabling downstream models to retain contextual coherence and support more complex reasoning tasks. This makes the method particularly advantageous in interactive systems where conversational quality is essential. However, it is inherently vulnerable to inference and re-identification attacks, as contextual cues may still allow adversaries to approximate original identities. Furthermore, maintaining consistent entity substitution across documents requires careful design to avoid semantic drift. This approach is therefore best suited to customer service workflows, general informational exchanges, and other low-risk environments where utility is prioritised over strict privacy guarantees.

XXXX Masking.

XXXX Masking replaces sensitive tokens with uniform placeholders (e.g., “XXXX”), offering one of the most privacy-preserving strategies by fully suppressing identifiable information. This technique benefits from high simplicity, minimal computational overhead, and robust protection against direct disclosure. However, it introduces substantial utility degradation, often resulting in fragmented or ambiguous text that undermines model interpretability and deteriorates user experience. Due to these drawbacks, this technique is most appropriate for high-security or compliance-driven applications—such as secure audit logs, redaction pipelines, or restricted archival systems—where privacy must take precedence over textual fidelity and usability.

LLM-based PII Removal.

LLM-based PII Removal leverages large language models to detect, classify, and rewrite sensitive content in a contextually informed manner. Its primary strengths include context-aware inference, enabling the system to identify PII even when phrasing is indirect, ambiguous, or domain-specific. Additionally, the method supports adaptive granularity, allowing for fine-tuned privacy depending on the application’s risk profile. Its balanced trade-off between data protection and semantic preservation positions it as

a highly versatile solution. Nonetheless, the technique is constrained by high computational requirements and potential variability in performance depending on the underlying model's quality and training corpus. Despite these limitations, LLM-based PII Removal is particularly well-suited to sensitive, high-value domains—including healthcare documentation, legal case processing, and financial advisory systems—where both privacy and linguistic quality are essential.

5.2.6 Comparative Summary

This section synthesises the findings from the evaluation of the three PII protection techniques—LLM-based PII Removal, Fake Names Replacement, and XXXX Masking—by comparing their relative strengths, limitations, and suitability across different operational contexts.

Overall Performance Ranking.

The comparative analysis indicates that LLM-based PII Removal provides the most favourable balance between privacy preservation and task utility, positioning it as the preferred method for general deployment. Fake Names Replacement demonstrates consistently high utility but only moderate privacy protection, making it well-suited for low-risk environments where data usefulness is prioritised. In contrast, XXXX Masking achieves the highest level of privacy but introduces significant degradation in downstream task performance, limiting its applicability to specialised, high-security scenarios.

Context-Dependent Recommendations.

The selection of an appropriate PII protection technique is inherently dependent on organisational requirements and contextual constraints. For scenarios demanding stringent privacy guarantees, methods such as XXXX Masking or LLM-based PII Removal are most appropriate. Environments prioritising maximal data utility, particularly where analytical accuracy is critical, benefit from Fake Names Replacement or LLM-based PII Removal. When projects require a balanced compromise between privacy and performance, LLM-based PII Removal remains the most effective option. Finally, for resource-constrained settings—including lightweight systems or operational

processes with limited computational capacity—both Fake Names Replacement and XXXX Masking offer viable, low-overhead alternatives.

Resource-Constrained Environments: Fake Names Replacement or XXXX Masking.

5.2.7 Limitations and Edge Cases

Despite promising experimental outcomes, several limitations and edge cases were observed across the evaluated PII protection methodologies. These factors must be considered when deploying the proposed framework in real-world systems.

Multi-Language Support.

The current implementation is primarily optimized for English-language inputs. As a result, performance may degrade when applied to linguistically diverse settings, particularly in languages with complex morphology, ambiguous entity boundaries, or low-resource characteristics. Extending the framework to multilingual environments would require additional training data, language-specific PII taxonomies, and potentially distinct model configurations.

Domain-Specific PII.

Several specialized domains—such as healthcare (e.g., ICD codes, lab identifiers), law (e.g., case references), and finance (e.g., transaction IDs)—contain highly domain-specific identifiers that generic NER or masking algorithms may fail to capture. These contexts demand tailored PII schemas and fine-tuned detection pipelines to ensure comprehensive protection.

Adversarial Inputs.

Sophisticated adversaries may intentionally obfuscate or encode sensitive information through misspellings, leetspeak, token fragmentation, or unconventional formatting. Such adversarial strategies may circumvent standard detection mechanisms and pose significant challenges to robust anonymization.

Temporal Consistency.

For long-form dialogues or multi-turn interactions, maintaining consistent encoding of entities (e.g., replacing the same name with the same alias across turns) becomes critical. Without effective state management, models may produce inconsistent substitutions, degrading both readability and contextual coherence.

Overall, the experimental results indicate that client-side PII encoding remains a viable strategy, offering a spectrum of privacy–utility trade-offs depending on the chosen method. Consequently, selecting an appropriate technique should be guided by application-specific requirements relating to privacy sensitivity, output quality, operational constraints, and computational resources.

The following section presents the final guidelines and the complete CSM Framework recommended on the basis of the experiments conducted in this study.

5.2.8 Final CSM Framework/ Guidelines

The following framework provides a decision-oriented representation of how privacy-preserving strategies can be selected based on contextual priorities such as privacy level, utility preservation, and resource availability. Each technique — *Fake Names Replacement*, *XXXX Masking*, and *LLM-based PII Removal* — is positioned according to its trade-offs between privacy, linguistic utility, and computational cost.

The framework begins with an assessment of application requirements, followed by a branching decision structure guiding the selection of the most appropriate anonymization method.

Figure 5.5 illustrates the CSM Framework Recommendations, highlighting the relative positioning and applicability of each sanitization approach.

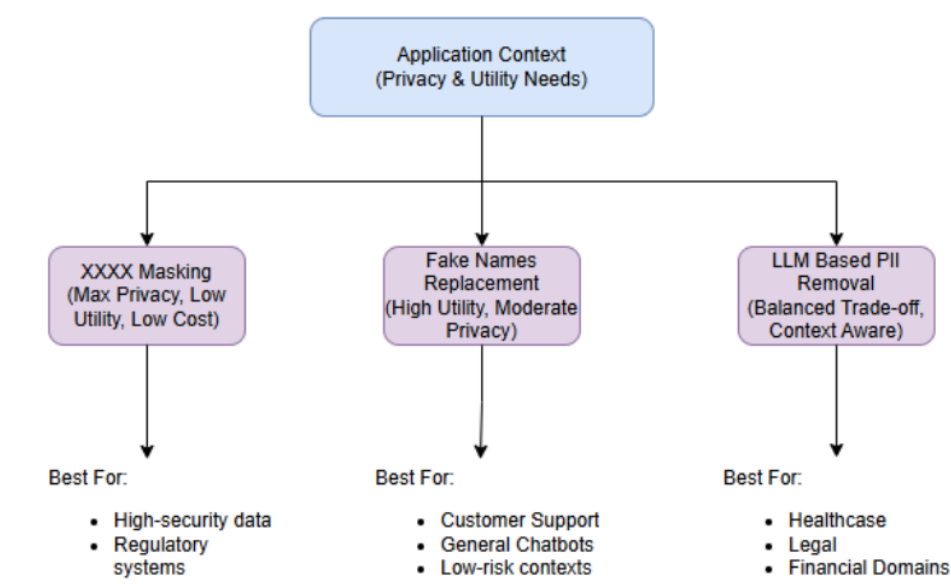


Figure 5.5 Framework Recommendations

Furthermore, Table 5.5 presents a comparative evaluation of the three methods in terms of privacy protection, utility preservation, computational cost, and recommended usage scenarios. This comparison provides a concise overview of how each technique performs across key evaluation dimensions.

Table 5.5 Experiments Comparison

Method	Privacy Level	Utility Retention	Computational Cost	Recommended Use
XXXX Masking	★★★★★	★★★☆☆	★★★★☆	High-security / strict compliance
Fake Names Replacement	★★★★☆	★★★★☆	★★★☆☆	General inquiries / low risk
LLM-based PII Removal	★★★★☆	★★★★☆	★★☆☆☆	Balanced / sensitive domains

CHAPTER 6

CONCLUSIONS AND RECOMMENDATIONS

6.1 Introduction

This chapter presents the key conclusions drawn from the research study on Privacy-Preserving Large Language Model (LLM) Inference. It also outlines the study’s contributions to knowledge and proposes future research directions.

The primary objective of this research was to develop and evaluate a privacy-preserving framework for LLM-based conversational systems. The CSM framework integrates various privacy transformation techniques—such as *real-name baseline*, *fake-name replacement*, *masking*, and *LLM-based PII removal*—and evaluates their impact on response quality using standardized metrics, namely BLEU, ROUGE, and Coherence Score.

Through rigorous experimentation, the study aimed to balance two critical objectives: safeguarding sensitive user information and maintaining the linguistic and semantic quality of generated outputs.

6.2 Discussion and Conclusion

The experimental analysis revealed that privacy-preserving mechanisms can be incorporated into LLM inference pipelines without causing significant degradation in text quality or coherence. Each privacy strategy exhibited unique trade-offs between privacy strength and output quality.

Trade-off Between Privacy and Linguistic Quality

Lightweight transformations such as *fake-name replacement* achieved strong privacy protection with minimal loss in fluency (as reflected in BLEU) and content preservation (as reflected in ROUGE). Conversely, aggressive masking strategies (e.g., replacing entities with “XXXX”) yielded higher privacy but reduced semantic richness.

Effectiveness of LLM-Based PII Removal

The use of an auxiliary LLM for entity redaction demonstrated balanced performance by maintaining sentence fluency and contextual integrity while effectively removing identifiable information.

Coherence Preservation

Despite entity-level substitutions, the Coherence Scores remained largely consistent across different configurations. This indicates that contextual continuity and logical flow can be preserved even after anonymization.

Framework Scalability and Adaptability

The designed pipeline supported batch processing and modular configuration, making it adaptable for multiple privacy settings and practical deployment in real-world conversational AI systems.

In summary, the study concludes that privacy-preserving LLM inference is achievable with minimal compromise to linguistic and semantic quality when appropriate anonymization techniques are applied judiciously. The research provides empirical validation for deploying such mechanisms in privacy-sensitive applications and provide a guidance to select the right methodology to prevent PII data with right level of performance.

6.3 Mapping of Objectives to Outcomes

Objective 1: Comprehensive Analysis of Existing Privacy-Preserving Methods

Outcome Achieved: A detailed literature review was conducted, covering cryptographic privacy mechanisms such as homomorphic encryption, secure multiparty computation, and differential privacy frameworks. The review highlighted the computational overhead and latency issues that limit their suitability for *real-time LLM inference* in sensitive domains (e.g., healthcare, banking). Identified gaps included the lack of lightweight, client-side methods capable of balancing privacy and usability, which guided the design of the proposed framework.

The study identified that current approaches often rely on server-side encryption or differential noise injection, which are unsuitable for real-time, user-facing chatbot

scenarios. These insights motivated the development of a *client-side NER-based anonymization approach -CSM Framework*.

Objective 2: Development of Client-Side PII Detection and Replacement Framework

Outcome Achieved: A client-side NER-based detection system was designed and implemented to identify PII entities (names, locations, organizations) from user inputs. The system supports dynamic replacement through synthetic fake-name substitution and masked anonymization (XXXX strategy), preserving sentence structure and semantic flow.

A context-aware replacement algorithm was developed to substitute detected entities with semantically consistent synthetic tokens (fake names or masked tokens), ensuring minimal coherence loss.

The framework achieved reliable detection accuracy and low average latency (measured in tokens/sec), validated across varying data inputs. The results confirmed feasibility for real-time privacy filtering prior to LLM inference.

Objective 3: Provide Framework to help in picking right methodology for balanced Utility and Performance Optimization.

Outcome Achieved: A modular, privacy-preserving inference CSM architecture was proposed and implemented, integrating the PII protection layer directly into the LLM pipeline. The architecture supports both real-time and batch processing modes, enabling flexible experimentation. Fig 5.5 and Table 5.5 provides that framework that let decide which methodology to pick in different use cases.

A comprehensive evaluation was conducted using BLEU, ROUGE, and Coherence Scores to assess linguistic quality and semantic preservation. Results demonstrated that privacy transformations achieved strong anonymization while maintaining acceptable coherence and fluency. The modular design also facilitated automated metric computation and scalability for large-scale deployment.

6.4 Contribution to Knowledge

This research advances the current understanding of privacy-preserving language model operations through several key contributions. First, it introduces a structured and modular evaluation framework designed to systematically assess the effects of privacy-

preserving transformations on large language model (LLM) outputs. This framework enables reproducible experimentation and supports robust comparative analysis across different anonymization techniques.

Second, the study incorporates a multidimensional evaluation strategy by integrating BLEU, ROUGE, and Coherence metrics. This combined metric approach allows for a comprehensive assessment of lexical similarity, semantic preservation, and contextual coherence, thereby offering a more holistic measure of text quality following privacy interventions.

Third, the research conducts an empirical assessment of multiple privacy mechanisms, including masking, fake-name substitution, and LLM-based redaction. The comparative analysis quantifies their respective impacts on output quality and coherence, providing a clearer understanding of the trade-offs inherent in different anonymization strategies. Furthermore, the work demonstrates the practical feasibility of employing client-side Named Entity Recognition (NER) for privacy preservation in real-time conversational systems. This contribution highlights the potential of lightweight, on-device processing for enhancing user data protection without compromising interaction quality.

Finally, the study establishes a reusable benchmark dataset and an accompanying evaluation methodology. These resources offer a standardized basis for future investigations in privacy-preserving text generation, facilitating greater consistency and comparability in subsequent research efforts.

6.5 Future Recommendations

Building on the findings and limitations of this study, several recommendations are proposed for future research and development:

- **Semantic-Aware Privacy Filters**

Future systems should incorporate advanced context-sensitive anonymization models capable of preserving the semantic integrity of sentences while providing robust entity-level privacy protection. Such models would enable more natural and coherent outputs even under stringent privacy constraints.

- **Differential Privacy and Federated Learning**

Integrating differential privacy techniques and federated learning architectures may offer mathematically provable privacy guarantees. These approaches can

ensure that sensitive user information remains secure, even when models are trained on distributed or client-side data sources.

- **Cross-Lingual and Multimodal Evaluation**

Extending the current framework to support multilingual datasets and multimodal inputs—such as text combined with speech or images—would significantly enhance its generalizability and applicability in diverse real-world settings.

- **User-Configurable Privacy Controls**

Providing end-users with customizable privacy settings can enable adaptable trade-offs between anonymity and output quality. This flexibility would allow users to select their preferred privacy level based on situational requirements and personal risk tolerance.

- **Human Evaluation Studies**

Providing end-users with customizable privacy settings can enable adaptable trade-offs between anonymity and output quality. This flexibility would allow users to select their preferred privacy level based on situational requirements and personal risk tolerance.

- **Industry-Level Integration**

The proposed pipeline may be further extended and deployed within commercial chatbot ecosystems—such as those in banking, healthcare, or customer support—to evaluate its performance under real-world constraints. Particular attention should be given to scalability, latency, and compliance with regulatory frameworks including the GDPR and India’s Digital Personal Data Protection (DPDP) Act.

6.6 Closing Remark

This research reaffirms that achieving privacy-preserving large language model inference is both technically feasible and practically valuable. The developed CSM framework offers a foundational approach for integrating privacy safeguards within modern AI systems.

CSM Framework also provided framework that help to pick right methodology for right use case.

By balancing privacy protection and linguistic quality, this study contributes toward the realization of responsible, trustworthy, and regulation-compliant AI — a necessary step for the safe adoption of generative language models in sensitive domains.

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APPENDIX A: RESEARCH PROPOSAL

PDF Link: [MS-GenAIChatbot_PII/Machine Learning/PII/Research_Proposal/Submission/Deeapkgoyal_ResearchProposal_PII_LLM.pdf](#)
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APPENDIX A: THESIS SOURCE CODE

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